

Highly Efficient, Regulated Dual-Output, Ambient Energy Manager for Source Voltage Level Configuration with Optional Primary Battery

Features

Constant input voltage regulation

- Optimized for constant voltage PV cells, intermittent and pulsed power.
- Input voltage up to 5 V with configurable regulation voltage from 50 mV to 4.5 V.
- Up to 110 mA current extracted from the harvester.

Ultra-low power start-up

- Cold start from 380 mV input voltage and 3 μ W input power (typical).

Integrated low-voltage LDO regulator

- Up to 20 mA load current.
- Output voltage configurable at 1.2 V or 1.8 V.
- Dynamically power-gated by external control.

Integrated high-voltage LDO regulator

- Up to 80 mA load current with 300 mV drop-out.
- Output voltage configurable within a range of 1.8 V to 4.1 V.
- Dynamically power-gated by external control.

Flexible energy storage management

- Configurable overcharge and over-discharge protection for any type of rechargeable storage elements.
- Fast supercapacitor charging.
- Status signal when storage element is running low and when LDO outputs are available.

Optional primary battery

- Automatic switching to primary battery when the storage element is depleted.

Integrated storage element balancing circuit for dual-cell supercapacitor

Applications

- Smart home
- Smart building
- Edge IoT
- Industrial sensor
- Retail
- PC accessories

Description

The AEM0094x is an integrated energy management circuit that extracts DC power from PV cells with constant MPP voltage, pulsed, intermittent, and capacitive sources to simultaneously store energy in a rechargeable element through its boost converter and supply the system with two independent regulated output voltages. The AEM0094x allows to extend battery lifetime and eliminate the primary energy storage element in a large range of applications.

With its unique cold-start circuit, it can start operating from 380 mV and only 3 μ W from the input with empty storage elements.

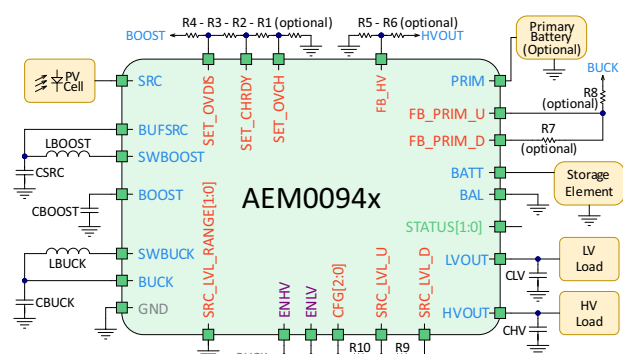
The low-voltage output typically drives a microcontroller while the high-voltage output typically drives a radio transceiver. Both are driven by highly-efficient LDO (Low Drop-Out) linear regulators for low noise and high stability.

Five capacitors, two inductors and two resistors are required, all available in small packages. With only nine external components, integration is maximized, footprint and BOM are minimized, optimizing the time-to-market and design costs.

The AEM00941 product variant allows higher efficiencies using a larger inductor value (typ. 100 μ H), while the AEM00940 allows higher current capabilities and smaller PCB size with a smaller inductor value (typ. 22 μ H).

Device Information

Description	Part number
AEM00940 - QFN 28-pin 5x5mm	10AEM00940C0000
AEM00941 - QFN 28-pin 4x4mm	10AEM00941C0000
AEM00940 - Evaluation board	2AAEM00940C001
AEM00941 - Evaluation board	2AAEM00941C001





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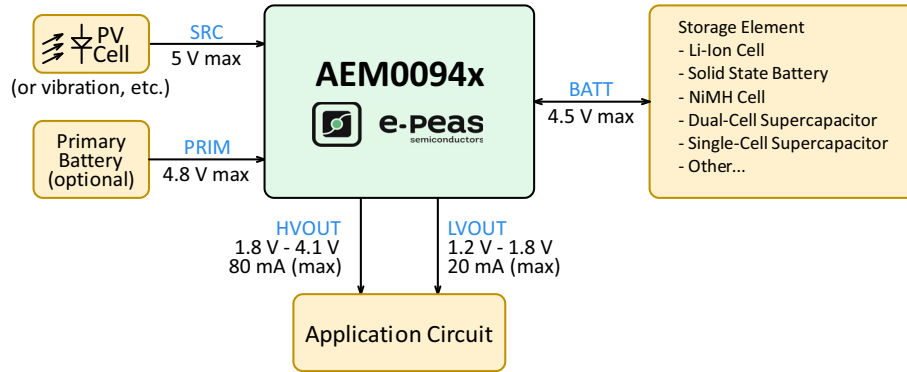


Figure 1: Simplified schematic view

1. Introduction

The AEM0094x is a full-featured energy efficient power management circuit capable of charging a storage element (battery or supercapacitor, connected to **BATT**) from an energy source (connected to **SRC**) as well as supplying loads at different operating voltages through two power supplying LDO regulators (**LVOUT** and **HVOUT**).

The heart of the AEM0094x is a cascade of two regulated switching converters, namely the boost converter and the buck converter, both with high power conversion efficiencies (See Section 3.6).

At first start-up, as soon as a required cold-start voltage of 380 mV and a scant amount of power of only 3 μ W are available from the harvested energy source, the AEM coldstarts. After the cold start, the AEM can extract the power available from the source and regulate the source at the voltage configured by the user.

Through three configuration pins (**CFG[2:0]**), the user can select a specific operating mode from a range of seven modes that covers most application requirements without any dedicated external component. These operating modes define the LDO output voltages and the storage element protection thresholds. A custom mode allows the user to define arbitrary storage element protection levels and the output voltage of the high-voltage LDO (See Section 6.1).

The source regulation voltage $V_{\text{SRC,REG}}$ can be adapted to the harvester thanks to two resistors and two configuration pins (**SRC_LVL_RANGE[1:0]**).

Two logic control pins (**ENLV** and **ENHV**) allow to dynamically activate or deactivate the LDO regulators that supply the low and high voltage loads. The status pin **STATUS[0]** alerts the user that the LDOs are operational and can be enabled. This signal can also be used to enable an optional external regulator.

If the storage element voltage gets depleted, **LVOUT** and **HVOUT** are power-gated and the controller is no longer supplied by the storage element to protect it from further discharge. Around 600 ms before the shutdown of the AEM, the status pin **STATUS[1]** alerts the user for a clean shutdown of the system.

However, if the storage element gets depleted and an optional primary battery is connected to **PRIM**, the AEM0094x automatically uses it as a source to recharge the storage element before switching back to the ambient source connected to **SRC**. This guarantees continuous operation even under the most adverse conditions (See Section 5.2.4). **STATUS[1]** is asserted when the primary battery is providing power.

2. Pin Configuration and Functions

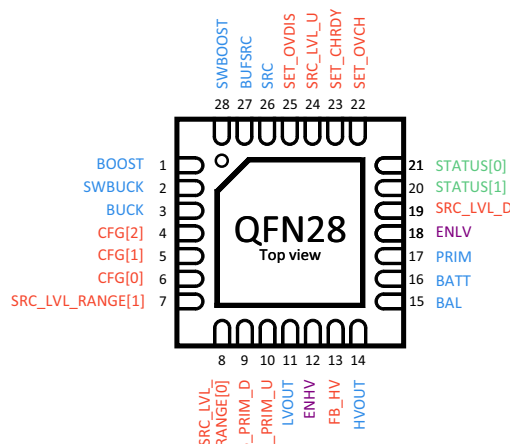


Figure 2: Pinout diagram QFN 28-pin

Name	Pin	Function
Power pins		
BOOST	1	Output of the boost converter.
SWBUCK	2	Switching node of the buck converter.
BUCK	3	Output of the buck converter.
LVOUT	11	Output of the low voltage LDO regulator.
HVOUT	14	Output of the high voltage LDO regulator.
BAL	15	Connection to the mid-point of a dual-cell supercapacitor (optional). Must be connected to GND if not used.
BATT	16	Connection to the energy storage element, battery or capacitor. Cannot be left floating.
PRIM	17	Connection to the primary battery (optional). Must be connected to GND if not used.
SRC	26	Connection to the harvested energy source.
BUFSRC	27	Connection to an external capacitor buffering the boost converter input.
SWBOOST	28	Switching node of the boost converter.

Table 1: Pins description (part 1)



Name	Pin	Logic Level		Function
		LOW	HIGH	
Configuration pins				
CFG[2]	4	GND	BUCK	Used to configure the storage element protection thresholds and the LDOs output voltages.
CFG[1]	5	GND	BUCK	
CFG[0]	6	GND	BUCK	
SRC_LVL_RANGE[1]	7	GND	BUCK	Used to configure the range for the source voltage regulation (see Section 6.2). Cannot be left floating.
SRC_LVL_RANGE[0]	8	GND	BUCK	
SRC_LVL_D	19	Analog Pin		Used to configure the source voltage regulation. Cannot be left floating.
SRC_LVL_U	24	Analog Pin		
FB_PRIM_D	9	Analog Pin		Used to configure the primary battery overdischarge voltage (optional). Must be connected to GND if not used.
FB_PRIM_U	10	Analog Pin		
FB_HV	13	Analog Pin		Used to configure the high-voltage LDO when in custom mode (optional). Must be left floating if not used.
SET_OVCH	22	Analog Pin		Used to configure the storage element protection thresholds when in custom mode (optional). Must be connected to BUCK if not used.
SET_CHRDY	23			
SET_OVDIS	25			
Control pins				
ENHV	12	GND	BUCK	Enabling pin for the high-voltage LDO (See Table 8).
ENLV	18	GND	BUCK to BOOST	Enabling pin for the low-voltage LDO (See Table 8). HIGH level is any voltage between BUCK and BOOST.
Status pins				
STATUS[1]	20	GND	BATT	Logic output. - HIGH: - during T _{CRIT} when in SHUTDOWN MODE. - as long as in PRIMARY BATTERY MODE. - LOW otherwise.
STATUS[0]	21	GND	BATT	Logic output. Asserted when the LDOs can be enabled.
Other pins				
GND	Exposed Pad		Ground connection, must be strongly tied to the PCB ground plane.	

Table 2: Pins description (part 2)

3. Specifications

3.1. Absolute Maximum Ratings

Parameter		Min	Max	Unit
Operating junction temperature T_j		-40	85	°C
Storage temperature T_{stg}		-65	150	°C
Input voltage	SRC, BUF SRC, BATT, BAL, PRIM, BOOST, SWBOOST, HVOUT, ENLV	-0.3	5.50	V
	BUCK, SWBUCK, LVOUT, CFG[2:0], FB_PRIM_U, FB_PRIM_D, SRC_LVL_RANGE[1:0], SRC_LVL_U, SRC_LVL_D, SET_OVDIS, SET_CHRDY, SET_OVCH, FB_HV, ENHV	-0.3	2.75	V

Table 3: Absolute maximum ratings

3.2. ESD Ratings

Parameter		Value	Unit
Electrostatic discharge V_{ESD}	Human-Body Model (HBM) ¹	± 1000	V
	Charged-Device Model (CDM) ²	± 1000	V

Table 4: Absolute maximum ratings

1. ESD Human-Body Model (HBM) value tested according to JEDEC standard JS-001-2023.

2. ESD charged-Device Mode (CDM) value tested according to JEDEC standard JS-002-2022.

ESD CAUTION	
	ESD (ELECTROSTATIC DISCHARGE) SENSITIVE DEVICE These devices have limited built-in ESD protection and damage may thus occur on devices subjected to high-energy ESD. Therefore, proper ESD precautions should be taken to avoid performance degradation or loss of functionality

3.3. Thermal Resistance

Package	Parameter	Value	Unit
QFN 28-pin 5 x 5 mm	Junction-to-Ambient thermal resistance (θ_{JA})	38.3	°C/W
	Junction-to-Case thermal resistance (θ_{JC})	2.183	°C/W
QFN 28-pin 4 x 4 mm	Junction-to-Ambient thermal resistance (θ_{JA})	60	°C/W
	Junction-to-Case thermal resistance (θ_{JC})	6	°C/W

Table 5: Thermal data



3.4. Electrical Characteristics at 25 °C

Symbol	Parameter	Condition	Min	Typ.	Max	Unit
Input voltage and input power						
P _{SRC,CS}	Minimum source power required to coldstart.			3		μW
V _{SRC,CS}	Minimum source voltage required to coldstart.			380		mV
V _{SRC}	Input voltage of the energy source (maximum given by the open-circuit voltage).	After cold start.	0.05 ¹		5	V
V _{SRC,REG}	Target regulation voltage of the source.		0.05		4.5	V
I _{SRC}	Harvested current from the energy source.	AEM00940: L _{BOOST} = 10 μH			110	mA
		AEM00940: L _{BOOST} = 22 μH			50	
		AEM00941: L _{BOOST} = 100 μH			10	
DC-DC converters						
V _{BOOST}	Output voltage of the boost converter.	During normal operation.	2.2		4.5	V
V _{BUCK}	Output voltage of the buck converter.		2	2.2	2.5	
V _{BUCK,RESET}	Minimum voltage on BUCK before switching to DEEP SLEEP MODE (from any other mode).			1.8		V
I _{BUCK}	Total load current supplied by the BUCK converter (including LVOUT current I _{LV}).	L _{BUCK} = 10 μH.	0		20	mA
		L _{BUCK} = 4 μH.	0		50	mA
Storage element						
V _{BATT}	Voltage on the storage element.		0 ²		4.5	V
V _{BAL,MIN}	Minimum voltage required on the BAL pin for the balancing circuit to operate.			0.9		V
V _{PRIM}	Voltage on the primary battery.		0.6		V _{OVDIS} + 0.4 V	V
I _{PRIM}	Current from the primary battery.	AEM00940: L _{BOOST} = 22 μH		20		mA
V _{FB_PRIM_U}	Feedback for defining the overdischarge voltage level on the primary battery.		0.15		1.1	V
V _{SRC_LVL_U}	Source feedback voltage on SRC_LVL_U pin.		0		1.35	V
V _{OVCH}	Maximum voltage accepted on the storage element before disabling the boost converter.	See Table 9.	2.30		4.50	V
V _{CHRDY}	Minimum voltage required on the storage element before enabling the LDO when coming from WAKE-UP MODE.	After cold start See Table 9.	2.250		4.45	V
V _{OVDIS}	Minimum voltage accepted on the storage element before switching to primary battery or entering SHUTDOWN MODE.	See Table 9.	2.20		4.40	V

Table 6: Electrical characteristics (Part 1)



Symbol	Parameter	Condition	Min	Typ.	Max	Unit
I _Q	Quiescent current on BATT when the boost converter is not running.	V _{BATT} = 3 V LDOs disabled.		400		nA
		V _{BATT} = 3 V LDOs enabled.		600		nA
Low-Voltage LDO regulator						
V _{LV} ³	Output voltage of the low-voltage LDO (see Table 9).		1.2		1.8	V
I _{LV}	Load current supplied by the low-voltage LDO.		0		20	mA
High-Voltage LDO regulator						
V _{HV} ⁴	Output voltage of the high-voltage LDO (see Table 9).		1.8		V _{OVDIS} - 0.3 V	V
I _{HV}	Load current supplied by the high-voltage LDO.		0		80	mA
Timing						
T _{VSRC,REG,UPDATE}	Time between two updates of the source voltage regulation (see Section 5.3).			20		s
T _{CRIT}	Time before shutdown once STATUS[1] has been asserted (see Section 5.2.5 and Figure 14).		400	600	800	ms

Table 6: Electrical characteristics (Part 2)

1. Minimum V_{SRC} value for harvesting capabilities after coldstart.
2. To stay in **NORMAL MODE**, V_{BATT} minimum voltage must stay above V_{OVDIS} .
3. The variability of V_{LV} at 1 mA is 1% (typical and preliminary result from simulations).
4. The variability of V_{HV} at 1 mA is 1.3% (typical and preliminary result from simulations).



3.5. Recommended Operating Conditions

Symbol	Parameter	Min	Typ.	Max	Unit
External components					
C_{SRC}	BUFSRC pin decoupling capacitor.	8	10	22	μF
C_{BOOST}	Output capacitor of the boost converter.	10	22		μF
L_{BOOST}	Inductor of the boost converter.	AEM00940	4	22 ¹	μH
		AEM00941		100	μH
C_{BUCK}	Output capacitor of the buck converter.	8	10		μF
L_{BUCK}	Inductor of the buck converter.	4	10	25	μH
C_{LV}	Low-voltage LDO regulator decoupling capacitor.	8	10	14	μF
C_{HV}	High-voltage LDO regulator decoupling capacitor.	8	10	14	μF
C_{BATT}	Capacitor connected on BATT if no storage element is connected (see Section 3.5.3).	LDOs disabled.	22		μF
		LDOs enabled.	150		μF
R_T	Optional - Sum of resistors for setting battery protection threshold voltages in custom mode. $R_T = R1 + R2 + R3 + R4$ (see Section 6.1).	1	10	100	$M\Omega$
R_V	Optional - Sum of resistors for setting the output voltage of the high-voltage LDO in custom mode. $R_V = R5 + R6$ (see Section 6.1).	1	10	40	$M\Omega$
R_P	Optional - Sum of resistors used to define the primary battery minimum voltage. $R_P = R7 + R8$ (see Section 6.3).	100		500	$k\Omega$
R_S	Sum of resistors used to define the source voltage regulation. $R_S = R9 + R10$ (see Section 6.2).	0.1		1	$M\Omega$

Table 7: Recommended operating conditions

1. For optimal efficiency, a 22 μH inductor is recommended on LBOOST. However, a 10 μH inductor may also be used as an alternative when higher source current is required, though this will result in slightly lower efficiency.

3.5.1. External Inductors Information

The AEM0094x operates with two external inductors. All inductors must support a minimum switching frequency of 10 MHz. Using inductors with low equivalent series resistance (ESR) improves the power-conversion efficiency of both the boost and buck converters.

L_{BOOST}

The AEM0094x circuit is typically implemented with one of the following values on L_{BOOST} :

- AEM00940:
 - 22 μH (peak current min. 115 mA) allows better efficiencies, especially at low SRC voltages.
 - 10 μH (peak current min. 250 mA) allows higher current from SRC to BATT.
- AEM00941: 100 μH (peak current min. 25mA) allows for higher efficiency at the expense of lower current driving capability.

L_{BUCK}

With the recommended operating condition (10 μH inductor), the buck inductor L_{BUCK} must support a minimum peak current of 50 mA.



3.5.2. External Capacitors Information

The AEM0094x operates with 6 external miniature capacitors to ensure stable operation of the boost converter, buck converter, storage element output, and LDO outputs. Each capacitor serves as a local energy buffer that limits voltage fluctuations caused by switching activity or dynamic load transition.

To maintain optimal performances and minimized quiescent current, all capacitors must exhibit a low leakage current and follow the recommended nominal values listed in Table 7, with a tolerance of $\pm 20\%$.

3.5.3. Storage Element Information

The energy storage element of the AEM0094x can be a rechargeable battery, a supercapacitor or a large capacitor. It should be selected so that its voltage never fall below V_{OVDIS} even during occasional peaks of the load current.

Connecting a decoupling capacitor (C_{BATT}) on the **BATT** pin is needed if:

- The internal resistance of the storage element cannot sustain V_{OVDIS} with the load current peaks;
- The application expects a disconnection of the storage element (e.g., because of a user removable connector, or during manufacturing), as the **BATT** pin cannot be left floating;
- The application does not require a storage element (see Section 6.4).

If the AEM0094x is used with no storage element or with a high-ESR storage element, the minimum C_{BATT} value required for optimal operation without voltage drops below V_{OVDIS} is:

- 150 μF if the LDOs are used.
- 22 μF if the LDOs are not used.



3.6. Typical Characteristics

3.6.1. Boost Conversion Efficiency for LBOOST = 22 μ H

The following graph shows the boost converter efficiency from SRC to BATT, with $L_{BOOST} = 22 \mu\text{H}$ (Coilcraft LPS4018-223MRB), including the loss of efficiency due to the AEM00940 quiescent current.

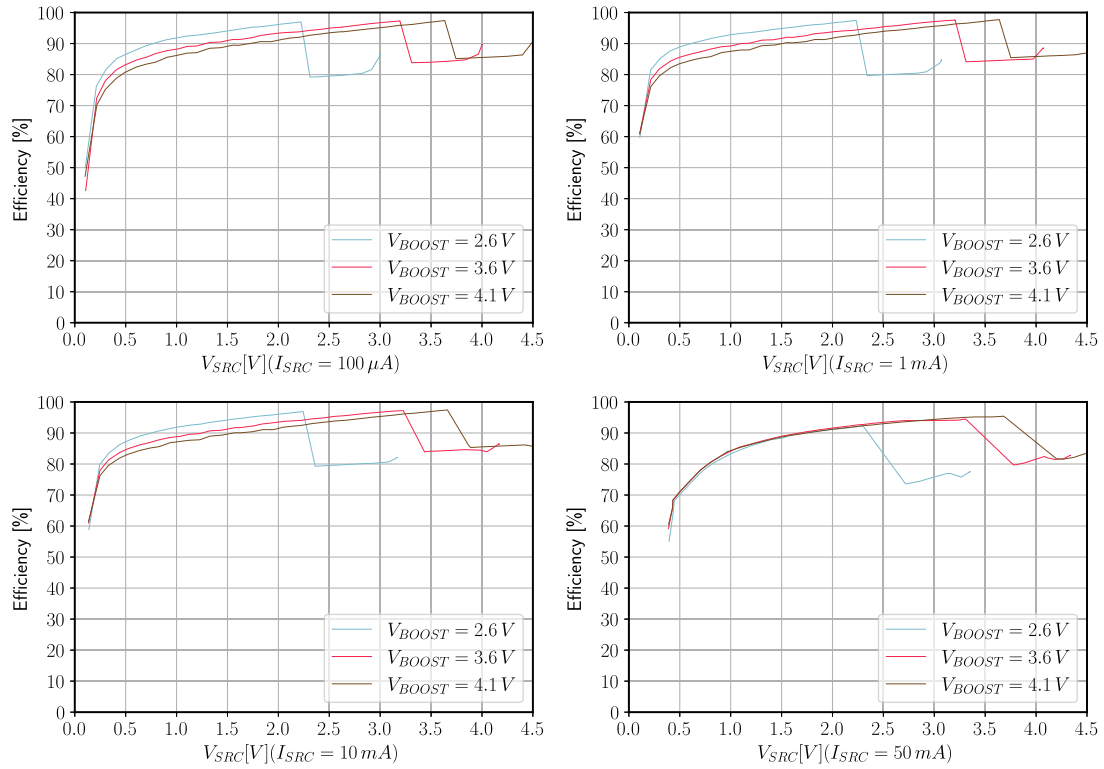


Figure 3: AEM00940 boost converter efficiency with $L_{BOOST} = 22 \mu\text{H}$ (Coilcraft LPS4018-223MRB)



3.6.2. Boost Conversion Efficiency for $L_{BOOST} = 10 \mu H$

The following graph shows the boost converter efficiency from SRC to BATT, with $L_{BOOST} = 10 \mu H$ (Coilcraft LPS4018-103MRB), including the loss of efficiency due to the AEM00940 quiescent current.

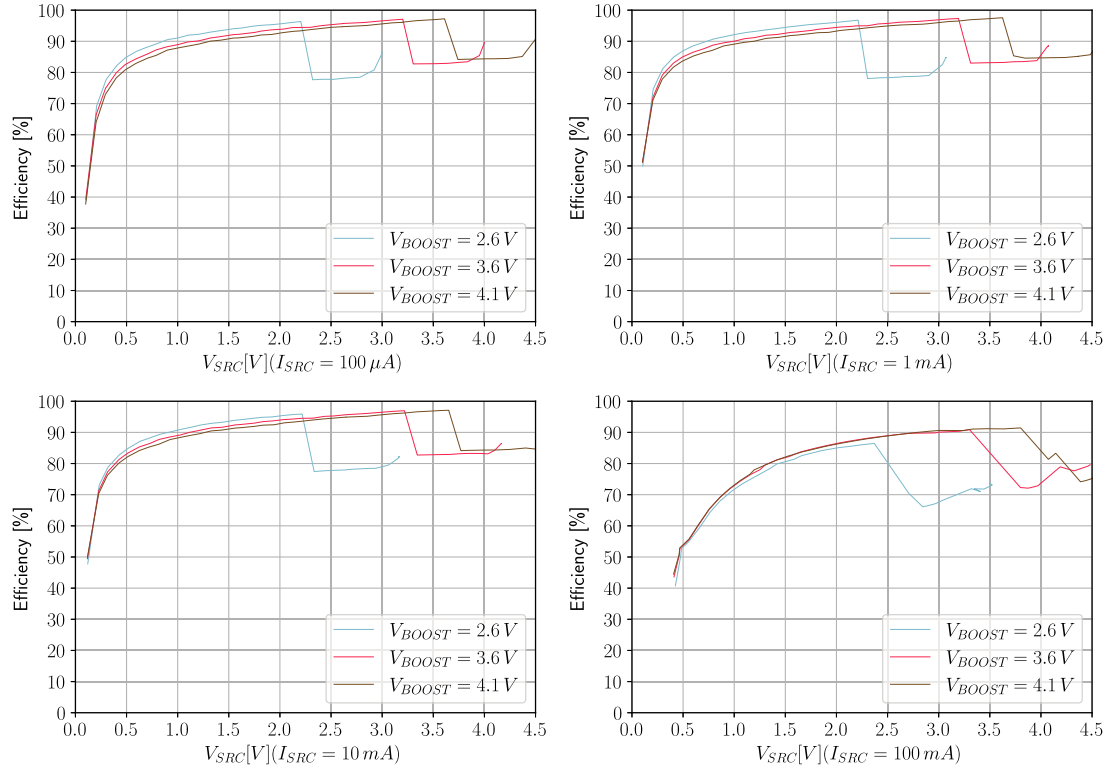


Figure 4: AEM00940 boost converter efficiency with $L_{BOOST} = 10 \mu H$ (Coilcraft LPS4018-103MRB)



3.6.3. Boost Conversion Efficiency for LBOOST = 100 μ H

The following graph shows the boost converter efficiency from SRC to BATT, with $L_{BOOST} = 100 \mu\text{H}$ (Coilcraft LPS5030-104MRC), including the loss of efficiency due to the AEM00941 quiescent current.

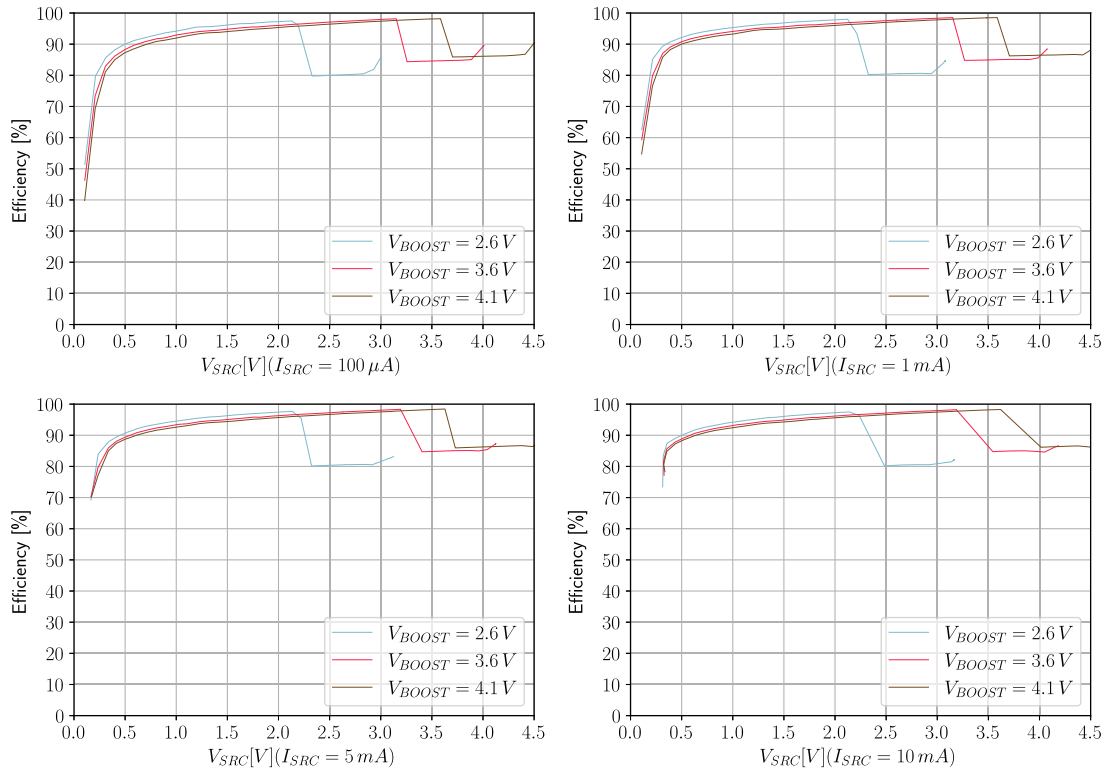


Figure 5: AEM00941 boost converter efficiency with $L_{BOOST} = 100 \mu\text{H}$ (Coilcraft LPS5030-104MRC)

3.6.4. Buck Conversion Efficiency

The following graph shows the buck converter efficiency from **BATT** to **BUCK**, with $L_{BUCK} = 10 \mu\text{H}$ (TDK MLZ1608N100LT000), with the AEM0094x quiescent current I_Q subtracted.

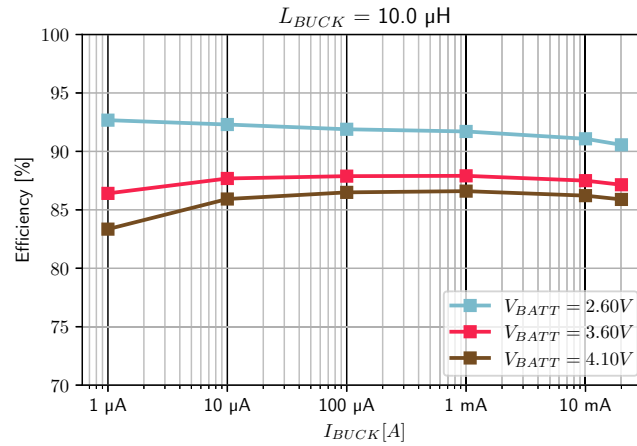


Figure 6: Buck converter efficiency with $L_{BUCK} = 10 \mu\text{H}$ (TDK MLZ1608N100LT000)

3.6.5. Quiescent Current

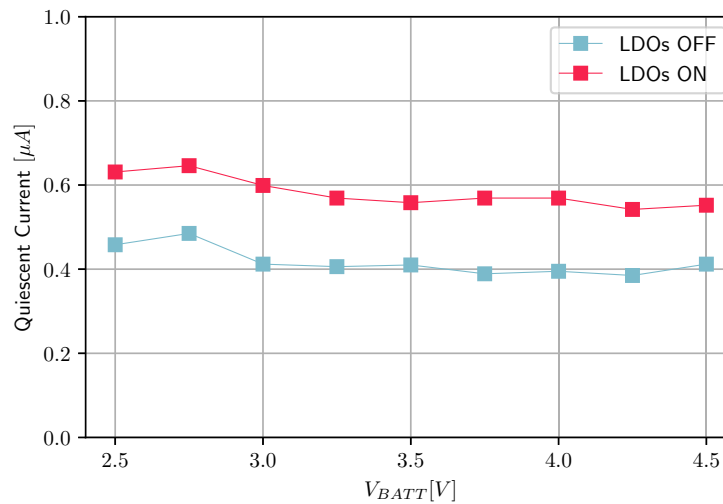


Figure 7: Quiescent current with LDOs on and off



3.6.6. High-voltage LDO Regulation

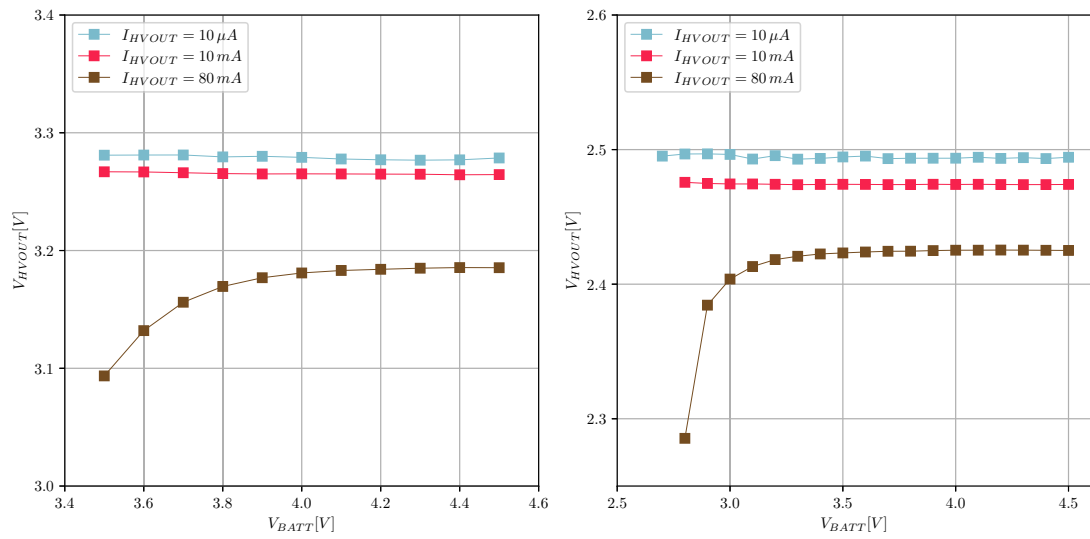


Figure 8: HVOUT at 3.3 V and 2.5 V

3.6.7. Low-voltage LDO Regulation

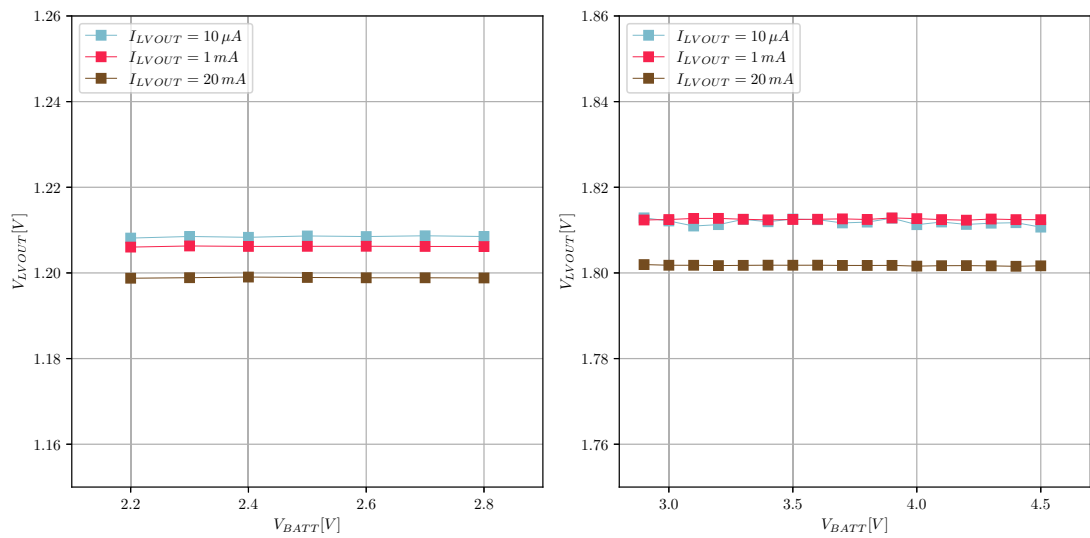


Figure 9: LVOUT at 1.2 V and 1.8 V

3.6.8. High-voltage LDO Efficiency

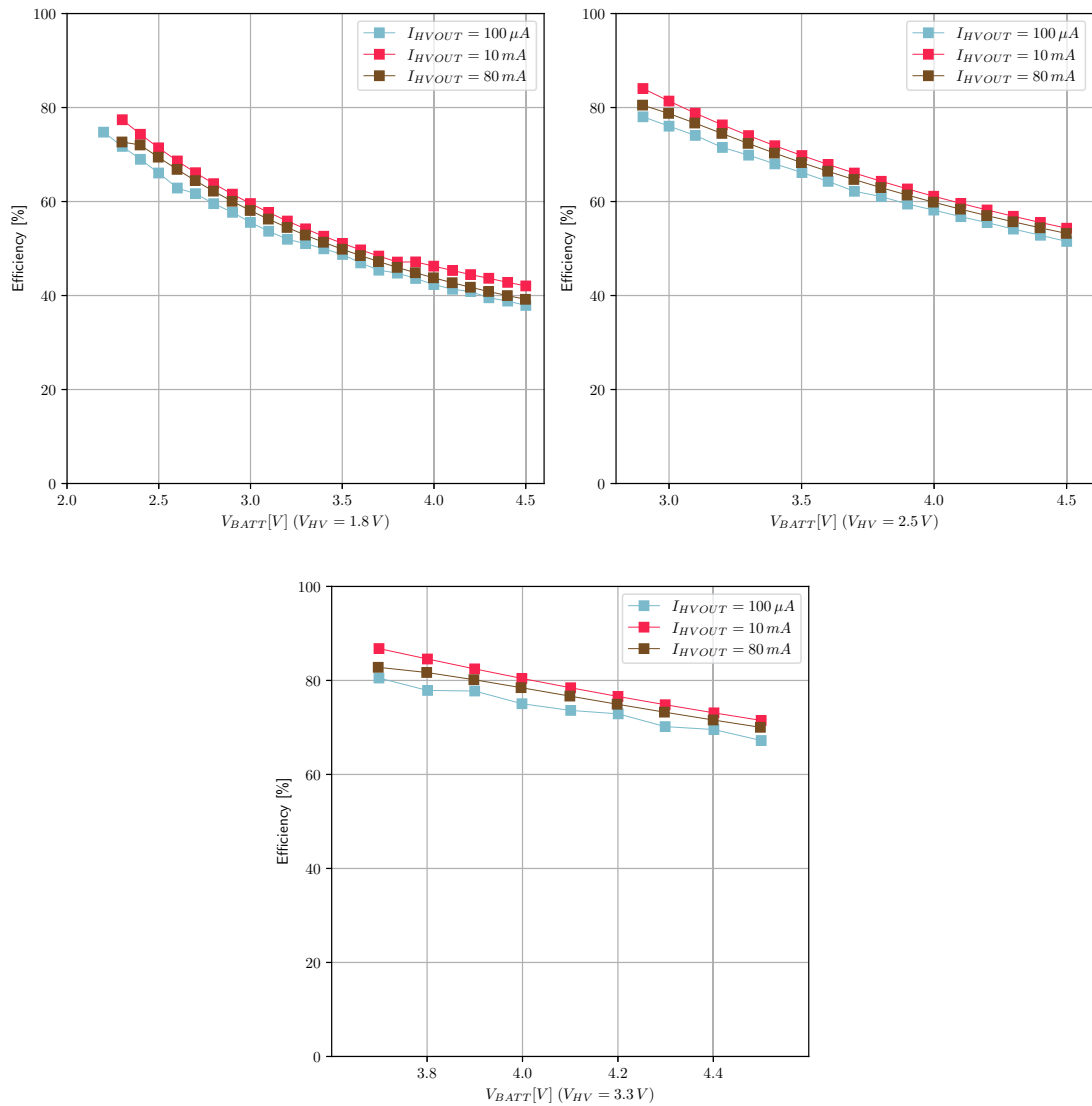


Figure 10: HVOUT efficiency at 1.8V, 2.5V and 3.3 V

The theoretical efficiency of an LDO can be calculated as V_{out} / V_{in} if quiescent current can be neglected with regards to the output current. For the high-voltage LDO, the theoretical efficiency is equal to V_{HV} / V_{BATT} .

3.6.9. Low-voltage LDO Efficiency

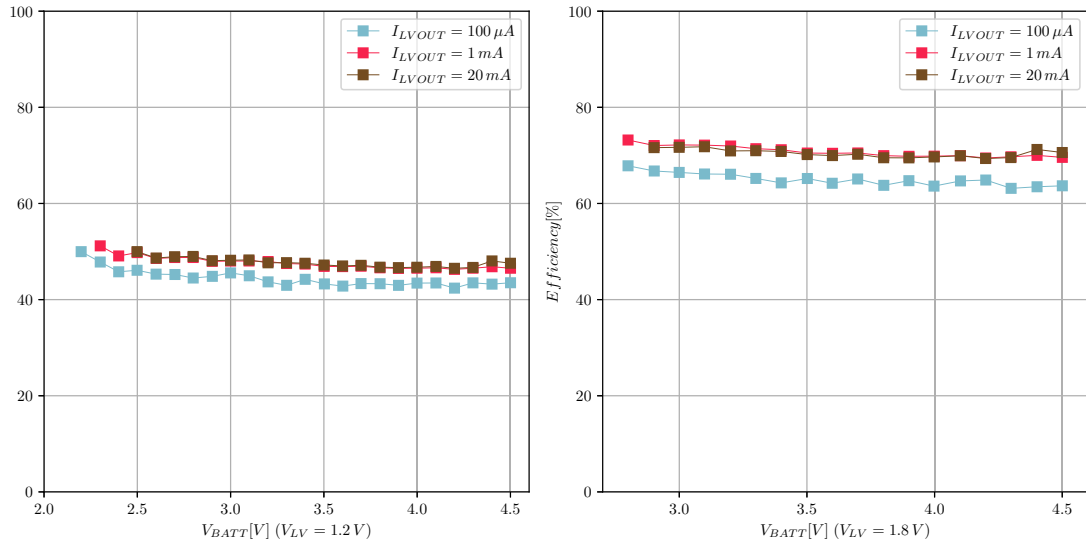


Figure 11: Efficiency of buck cascaded with LVOUT at 1.2 V and 1.8 V

The theoretical efficiency of an LDO can be calculated as V_{LV} / V_{BUCK} . Starting from the battery, the efficiency of the buck converter (η_{BUCK}) has to be taken into account (see Figure 13).

The efficiency between V_{BATT} and V_{LV} is therefore equal to:

$$\eta_{BUCK} \cdot \frac{V_{LV}}{V_{BUCK}}$$



4. Functional Block Diagram

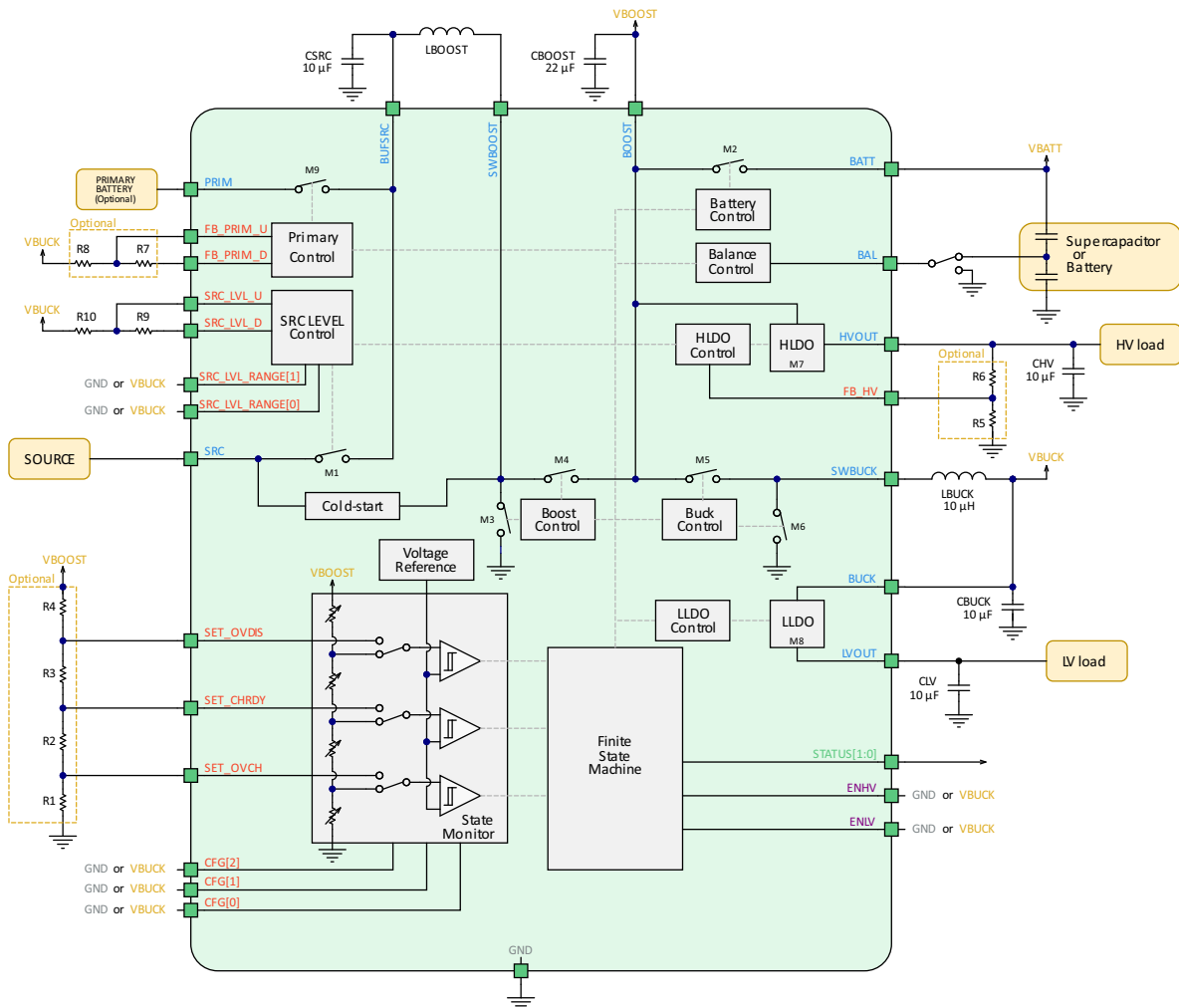


Figure 12: Functional block diagram

5. Theory of Operation

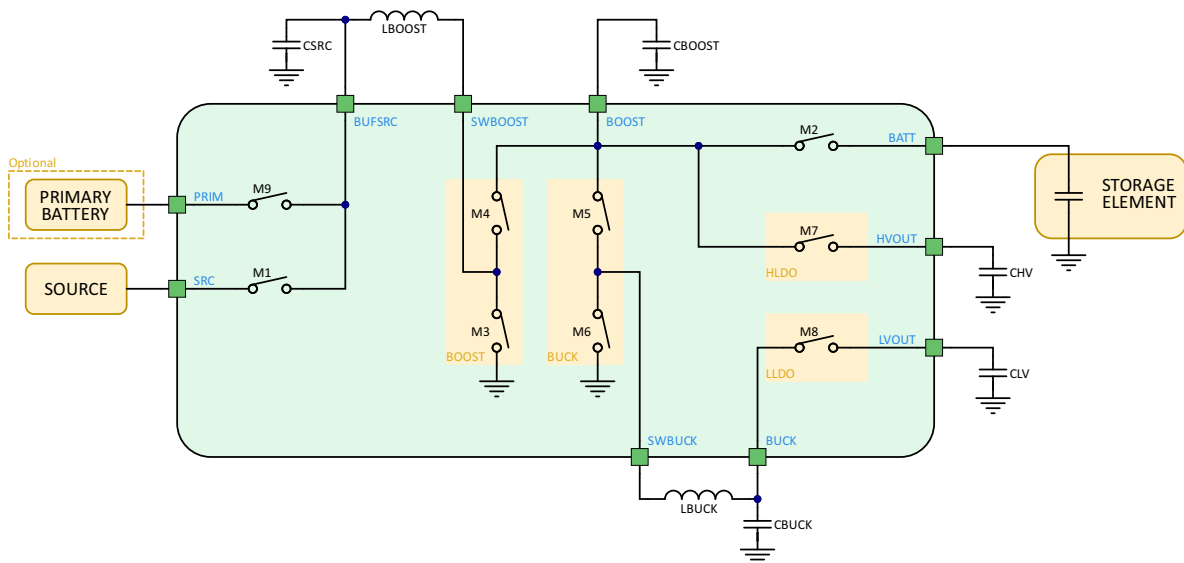


Figure 13: Simplified schematic view of the AEM0094x

5.1. Power Converters

5.1.1. Boost Converter

The boost (or step-up) converter raises the voltage available at **BUFSRC** to a level suitable for charging the storage element, in the range of 2.2 V to 4.5 V, according to the system configuration. This voltage (V_{BOOST}) is available at the **BOOST** pin.

The switching transistors of the boost converter are M3 and M4, with the switching node available externally at **SWBOOST**. The reactive power components of this converter are the external inductor L_{BOOST} and the external capacitor C_{BOOST} .

The regulation control circuit periodically disconnects the source from **BUFSRC** pin with the transistor M1 in order to measure the voltage present on **SRC_LVL_U** and define the source regulation voltage.

BUFSRC is decoupled by the capacitor C_{SRC} , which smooths the voltage against the current pulses induced by the boost converter.

The storage element is connected to the **BATT** pin. Its voltage is named V_{BATT} . This node is linked to **BOOST** through the transistor M2. In **NORMAL MODE** (see Section 5.2.2), this transistor effectively shorts the battery to the **BOOST** node ($V_{\text{BATT}} = V_{\text{BOOST}}$). When energy harvesting is occurring, the boost converter delivers a current that is shared between the battery and the LDOs. M2 is opened to disconnect the storage element when V_{BATT} reaches V_{OVDIS} . However, in such a scenario, the AEM0094x offers the possibility of connecting a primary battery to recharge V_{BATT} up to V_{CHRDY} . In this case, the transistor M9 connects **PRIM** to **BUFSRC** and the transistor M1 is opened to disconnect the **SRC** input pin as explained in the **PRIMARY BATTERY MODE** section.

To ensure the boost converter operates correctly:

- when using **PRIM** as input of the boost converter (in **PRIMARY BATTERY MODE**), the voltage of the primary battery must remain below $V_{\text{OVDIS}} + 0.4$ V (see Table 6).
- when using **SRC** as input of the boost converter, the harvester maximum current must not exceed I_{SRC} maximum value (see Table 6).

More explanations about the different modes can be found in Section 5.2.



5.1.2. Buck Converter

The buck (or step-down) converter lowers the voltage from V_{BOOST} to a constant V_{BUCK} value of 2.2 V. This voltage is available at the **BUCK** pin. The switching transistors of the buck converter are M5 and M6, with the switching node available externally at **SWBUCK**. The reactive power components of the buck converter are the external inductor L_{BUCK} and the external capacitor C_{BUCK} .

5.1.3. LDO Outputs

Two Low Drop-Out linear regulators are available to supply loads at different operating voltages:

- Through M7, **BOOST** supplies the high-voltage LDO that powers its load through **HVOUT**. This regulator delivers a clean voltage named V_{HV} . When using the built-in configuration modes, an output voltage of 1.8 V, 2.5 V or 3.3 V can be configured through **CFG[2:0]**. When using the custom configuration mode, V_{HV} is adjustable between 1.8 V and $V_{OVDIS} - 0.3$ V. The output is decoupled by the external capacitor C_{HV} .
- Through M8, V_{BUCK} supplies the low-voltage LDO that powers its load through **LVOUT**. This regulator delivers a clean voltage named V_{LV} of 1.2 V or 1.8 V configured through **CFG[2:0]**. The output is decoupled by the external capacitor C_{LV} .

See Table 6 for **HVOUT** and **LVOUT** maximum current values (respectively I_{HV} and I_{LV}).

Both the high-voltage and the low-voltage outputs can be dynamically enabled or disabled respectively with the logic control pins **ENHV** and **ENLV** (see table below).

ENLV	LVOUT	ENHV	HVOUT
L	Disabled	L	Disabled
H	Enabled	H	Enabled

Table 8: LDOs configurations

5.2. Operating Modes

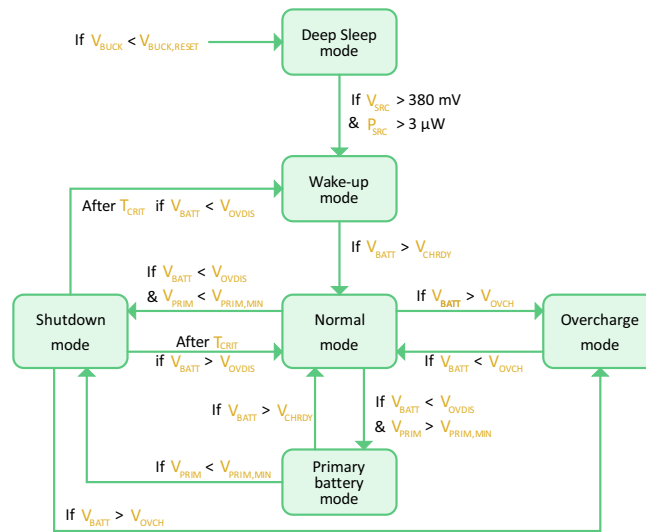


Figure 14: Diagram of the AEM0094x modes

5.2.1. Deep Sleep & Wake Up Modes

The **DEEP SLEEP MODE** is a state where all nodes are deeply discharged and there is no available energy to be harvested.

The AEM0094x enters **DEEP SLEEP MODE** if V_{BUCK} falls below $V_{BUCK,RESET}$ (see Table 6).

In this mode, as soon as the required cold-start voltage of 380 mV and the required power of 3 μ W becomes available on **SRC**, the **WAKE-UP MODE** is activated. V_{BOOST} and V_{BUCK} rise up to a voltage of 2.2 V. V_{BOOST} then rises up to V_{OVCH} .

At this stage, both LDOs are internally disabled. Therefore, **STATUS[0]** is low as shown in Figure 20 and Figure 21.

When V_{BOOST} reaches V_{OVCH} , two scenarios are possible:

- In the first scenario, a supercapacitor or a capacitor having a voltage lower than V_{CHRDY} is connected to the **BATT** node (see Section 5.2.1.1).
- In the second scenario, a charged battery is connected to the **BATT** node (see Section 5.2.1.2).

5.2.1.1. Supercapacitor as a Storage Element

If the storage element is a supercapacitor, the storage element may need to be charged from 0 V. The boost converter charges **BATT** from the input source and by modulating the conductance of M1 and M2. During the charge of the **BATT** node, both LDOs are disabled and **STATUS[0]** is low. When V_{BATT} reaches V_{CHRDY} , the circuit enters **NORMAL MODE**, **STATUS[0]** is asserted and the LDOs can be enabled by the user using **ENLV** and **ENHV** control pins as shown in Figure 20.

5.2.1.2. Battery as a Storage Element

If the storage element is a battery but its voltage is lower than V_{CHRDY} , the storage element first needs to be charged until it reaches V_{CHRDY} . This allows a safety margin to ensure that the storage element is able to provide the required power before enabling the outputs (LDOs).

Once V_{BATT} exceeds V_{CHRDY} , or if the battery was initially charged above V_{CHRDY} , the circuit enters **NORMAL MODE**. **STATUS[0]** is asserted and the LDOs can be dynamically enabled or disabled through **ENLV** and **ENHV** as shown in Figure 21.



5.2.2. Normal Mode

Once the AEM enters **NORMAL MODE**, it stays in this mode as long as the following condition is met:

$$V_{OVDIS} < V_{BATT} < V_{OVCH}$$

The AEM0094x will switch to another mode in the following cases:

- V_{BATT} increases above V_{OVCH} because the source provides more power than the load consumes. The circuit enters **OVERCHARGE MODE**, as explained in Section 5.2.3.
- V_{BATT} falls below V_{OVDIS} due to a lack of power from the source. In this case, either the circuit enters **SHUTDOWN MODE** as explained in Section 5.2.5, or, if a charged primary battery is connected on **PRIM**, the circuit enters **PRIMARY BATTERY MODE** as explained in Section 5.2.4.

5.2.3. Overcharge Mode

When V_{BATT} reaches V_{OVCH} , the battery charge is complete. The AEM maintains V_{BATT} around V_{OVCH} , with an hysteresis of a few mV as shown in Figure 22, to prevent damage to the storage element and to the internal circuitry. In this configuration, the boost converter is periodically activated to maintain V_{BATT} and the LDO output voltages are available. Moreover, when the boost converter is not activated, the transistor M1 in Figure 13 is opened to prevent current from the source to the storage element when V_{SRC} is higher than V_{OVCH} .

5.2.4. Primary Battery Mode

When V_{BATT} drops below V_{OVDIS} , the circuit compares the voltage on **PRIM** with the voltage on **FB_PRIM_U** to determine whether a charged primary battery is connected on **PRIM**. The voltage on **FB_PRIM_U** is set thanks to two optional resistors as explained in Section 6.3.

If the following formula is true, the circuit considers the primary battery as available and the circuit enters **PRIMARY BATTERY MODE**.

$$\frac{V_{PRIM}}{4} > V_{FB_PRIM_U}$$

In that mode, transistor M1 is opened and the primary battery is connected to **BUFSRC** through transistor M9 to become the source of energy of the AEM0094x. **STATUS[1]** is asserted as long as the chip is in **PRIMARY BATTERY MODE**.

The AEM remains in this mode until:

- V_{BATT} reaches V_{CHRDY} . At that point, the circuit enters **NORMAL MODE**.
- The primary battery is depleted (above formula becomes false). The circuit switches to **SHUTDOWN MODE**.

If no primary battery is used in the application, **PRIM**, **FB_PRIM_U** and **FB_PRIM_D** must be tied to **GND**.

5.2.5. Shutdown Mode

When V_{BATT} drops below V_{OVDIS} and no power is available from a primary battery, the circuit enters **SHUTDOWN MODE**, as shown in Figure 23, to prevent deep discharge that could damage the storage element and make the LDOs unstable. The circuit asserts **STATUS[1]** to warn the application that a shutdown may occur. Both LDO regulators remain enabled during T_{CRIT} (about 600 ms).

If no primary battery is used, this mechanism allows the application circuit, whether it is powered on **LVOUT** or **HVOUT**, to trigger an interrupt by the low to-high transition of **STATUS[1]**, and to take all appropriate actions before **LVOUT** and **HVOUT** are disabled.

After T_{CRIT} :

- If V_{BATT} has recovered above V_{OVDIS} , the AEM switches back to **NORMAL MODE**.
- If V_{BATT} has not recovered above V_{OVDIS} , the circuit enters **WAKE-UP MODE**. Both LDOs are disabled and **BATT** is disconnected from **BOOST** to avoid damaging the battery due to the over-discharge. From now on, the AEM must go through the wake-up procedure described in the Section 5.2.1.

If V_{BATT} reaches V_{OVCH} while in **SHUTDOWN MODE**, the AEM0094x switches directly to **OVERCHARGE MODE** without waiting the end of T_{CRIT} .

5.3. Source Voltage Regulation

During **NORMAL MODE**, **SHUTDOWN MODE** and a part of **WAKE-UP MODE**, the source voltage (V_{SRC_REG}) is regulated thanks to the evaluation of the voltage on the **SRC_LVL_U** pin and the gain set by **SRC_LVL_RANGE[1:0]** pins:

$$V_{SRC_REG} = V_{SRC_LVL_U} \cdot \text{Gain}$$

With the resistive divider shown in Figure 16, the AEM0094x measures $V_{SRC_LVL_U}$ by connecting **SRC_LVL_D** to GND internally. This evaluation is performed every $T_{VSRC_REG_UPDATE}$ (20 s).

$V_{SRC_LVL_U}$ is then multiplied by the gain to define the source voltage regulation. Three different gain values are available through the configuration pins **SRC_LVL_RANGE[1:0]** as shown in Table 10.

With the exception of this evaluation process, the source voltage V_{SRC} is continuously compared to V_{SRC_REG} :

- When V_{SRC} exceeds V_{SRC_REG} by a small hysteresis, the boost converter is switched on, extracting electric charges from the source, thus lowering its voltage.
- When V_{SRC} falls below V_{SRC_REG} by a small hysteresis, the boost converter is switched off, allowing the harvester to accumulate new electric charges into C_{SRC} , which voltage rises.

5.4. Storage Element Balancing Circuit for Dual-cell Supercapacitor

When using a dual-cell supercapacitor (with the cells in series) it is necessary to keep both cells at similar voltages to avoid damage due to a potential over-voltage on one cell. This is ensured by the AEM0094x storage element balancing circuit (**BAL** pin).

If a battery, a capacitor or a single-cell supercapacitor is connected on **BATT**, the balancing circuit is not needed. **BAL** must be connected to GND to disable the storage element balancing circuit.

If a dual-cell supercapacitor is connected on **BATT**, **BAL** must be connected to the node between the two cells of the supercapacitor. The storage element balancing circuit compensates for any mismatch of the two cells that could overcharge one of the two cells. It ensures that **BAL** voltage remains close to $V_{BATT} / 2$.

The balancing circuit works as follows:

- The balancing circuit is enabled only when $V_{BAL} > V_{BAL_MIN}$ (see Table 6).
- If $V_{BAL} > \frac{V_{BATT}}{2}$: the AEM0094x enables a switch between **BAL** and GND to momentarily discharge the bottom cell to GND until balance is restored.
- If $V_{BAL} < \frac{V_{BATT}}{2}$: the AEM0094x enables a switch between **BATT** and **BAL** to transfer charge from the top cell to the bottom cell until balance is restored.

6. System Configuration

6.1. Battery and LDOs Configuration

Configuration pins			Storage element threshold voltages			LDOs output voltages		Typical use
CFG [2]	CFG [1]	CFG [0]	V _{OVCH}	V _{CHRDY}	V _{OVDIS}	V _{HV}	V _{LV}	
H	H	H	4.12 V	3.67 V	3.60 V	3.3 V	1.8 V	Li-ion battery
H	H	L	4.12 V	4.04 V	3.60 V	3.3 V	1.8 V	Solid state battery
H	L	H	4.12 V	3.67 V	3.01 V	2.5 V	1.8 V	Li-ion/NiMH battery
H	L	L	2.70 V	2.30 V	2.20 V	1.8 V	1.2 V	Single-cell supercapacitor
L	H	H	4.50 V	3.67 V	2.80 V	2.5 V	1.8 V	Dual-cell supercapacitor
L	H	L	4.50 V	3.92 V	3.60 V	3.3 V	1.8 V	Dual-cell supercapacitor
L	L	H	3.63 V	3.10 V	2.80 V	2.5 V	1.8 V	LiFePO4 battery
L	L	L	Custom mode - Programmable with R1 to R6.				1.8 V	

Table 9: Usage of CFG[2:0]

DISCLAIMER: storage element thresholds provided in the table above are indicative to support a wide range of storage element variants. They are provided as is to the best knowledge of e-peas's application laboratory. They should not replace the actual values provided in the storage element manufacturer's specifications and datasheet.

Through three configuration pins (CFG[2:0]), the user can set a particular operating mode from a range that covers most application requirements, as shown in Table 9, without any dedicated external component. The three threshold levels are defined as:

- V_{OVCH}: maximum voltage accepted on the storage element before disabling the boost converter.
- V_{CHRDY}: minimum voltage required on the storage element after a cold start before enabling the LDOs.
- V_{OVDIS}: minimum voltage accepted on the storage element before considering the storage element as depleted.

See Section 5 for more information about the purposes of these thresholds.

The two LDOs output voltages are called V_{HV} and V_{LV} for the high and low output voltages respectively. Seven combinations of these voltage levels are hard-wired and selectable through the CFG[2:0] configuration pins, covering most application cases. For other V_{OVCH}, V_{CHRDY}, V_{OVDIS} and V_{HV} voltages combinations, a custom mode is available. In this mode, the user can define those voltages with resistors, connected to the pins named SET_OVDIS, SET_CHRDY, SET_OVCH and FB_HV.

When the custom mode is not used, SET_OVDIS, SET_CHRDY and SET_OVCH pins must be connected to V_{BUCK} and FB_HV must be left floating.

6.1.1. Custom Mode

When **CFG[2:0]** are tied to **GND**, the custom mode is selected. All six resistors, shown in Figure 15, are used to configure the custom mode as follows:

V_{OVCH} , V_{CHRDY} and V_{OVDIS} are defined thanks to **R1**, **R2**, **R3** and **R4**. The resistors are calculated as follows:

- $R_T = R1 + R2 + R3 + R4$
- $1M\Omega \leq R_T \leq 100M\Omega$
- $R1 = R_T \cdot \frac{1V}{V_{OVCH}}$
- $R2 = R_T \cdot \left(\frac{1V}{V_{CHRDY}} - \frac{1V}{V_{OVCH}} \right)$
- $R3 = R_T \cdot \left(\frac{1V}{V_{OVDIS}} - \frac{1V}{V_{CHRDY}} \right)$
- $R4 = R_T \cdot \left(1 - \frac{1V}{V_{OVDIS}} \right)$

V_{HV} is defined thanks to **R5** and **R6**. The resistive divider is configured as follows:

- $R_V = R5 + R6$
- $1M\Omega \leq R_V \leq 40M\Omega$
- $R5 = R_V \cdot \frac{1V}{V_{HV}}$
- $R6 = R_V \cdot \left(1 - \frac{1V}{V_{HV}} \right)$

NOTE: If **ENHV** = L (**HVOUT** is disabled), **R5** and **R6** are not needed, **FB_HV** should be left floating.

The resistors should have high values to make the current flowing through them negligible. Moreover, the following constraints must be met to ensure the functionality of the chip:

- $V_{CHRDY} + 0.05V \leq V_{OVCH} \leq 4.5V$
- $V_{OVDIS} + 0.05V \leq V_{CHRDY} \leq V_{OVCH} - 0.05V$
- $2.2V \leq V_{OVDIS}$
- $V_{HV} \leq V_{OVDIS} - 0.3V$

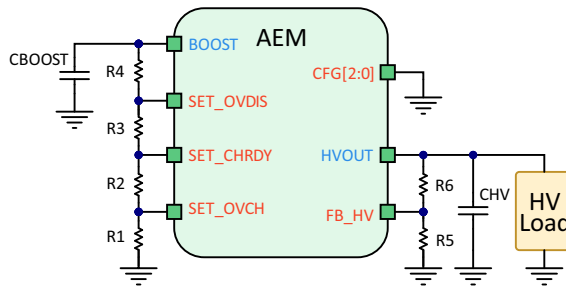


Figure 15: Custom configuration resistors

6.2. Source Voltage Configuration

To define V_{SRC_REG} , the user must configure:

- a resistive divider, as shown in Figure 16, calculated as follows:
 - $R_S = R9 + R10$
 - $100k\Omega \leq R_S \leq 1M\Omega$
 - $V_{SRC_LVL_U} = \frac{V_{SRC_REG}}{Gain}$
 - $R9 = \frac{V_{SRC_REG}}{Gain} \cdot R_S \cdot \frac{1}{2.2V}$
 - $R10 = R_S - R9$
- the **SRC_LVL_RANGE[1:0]** pins as shown in Table 10.

*NOTE: The **SRC_LVL_U** voltage measure is done thanks to an Analog-to-Digital Converter (ADC) with a 27.5 mV resolution. When the measured voltage falls between two values, the higher of the two values is used as $V_{SRC_LVL_U}$ for V_{SRC_REG} regulation.*

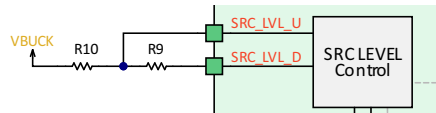


Figure 16: SRC_LVL_U and SRC_LVL_D connections

Configuration pins	Function	
SRC_LVL_RANGE[1:0]	Gain	V_{SRC_REG} range
LL	x1	$V_{SRC_REG} < 1.35\text{ V}$
LH	x2	$1.35\text{ V} < V_{SRC_REG} < 2.70\text{ V}$
HL	x4	$2.70\text{ V} < V_{SRC_REG} < 4.47\text{ V}$
HH		

Table 10: SRC_LVL_RANGE[1:0] configuration

6.3. Primary Battery Configuration

To use the primary battery, it is mandatory to determine V_{PRIM_MIN} , the voltage at which the primary battery is considered as fully depleted. The circuit uses a resistive divider between **BUCK** and **FB_PRIM_D** to define the voltage on **FB_PRIM_U** as V_{PRIM_MIN} divided by 4. During V_{PRIM_MIN} evaluation, the circuit connects **FB_PRIM_D** to **GND**. When V_{PRIM_MIN} is not evaluated, **FB_PRIM_D** is left floating to avoid quiescent current on the resistive divider. The resistors are calculated as follows:

- $R_P = R7 + R8$
- $100k\Omega \leq R_P \leq 500k\Omega$
- $R7 = \frac{V_{PRIM_MIN}}{4} \cdot R_P \cdot \frac{1}{2.2V}$
- $R8 = R_P - R7$

*NOTE: **FB_PRIM_U**, **FB_PRIM_D** and **PRIM** must be tied to **GND** if no primary battery is used.*

6.4. No-battery Configuration

If the application does not use a storage element, the PCB must include a capacitor on the **BATT** pin. See Section 3.5.3 for C_{BATT} value.

The storage element may not be necessary in the following cases:

- If the harvested energy source is permanently available and covers the application purposes.
- If the application does not need to store energy when the harvested energy source is not available.

6.5. Supplying an Application Circuit with BUCK

It is possible to supply an application circuit directly from **BUCK**, with the benefit of high **BATT** to **BUCK** efficiency, provided that the following conditions are met:

- The application circuit can be supplied from a voltage in the 2.0 V - 2.5 V range (V_{BUCK} is typically 2.2 V with ripple, see Table 6).
- The sum of the following currents must be below the maximum I_{BUCK} value (see Table 6):
 - Current of the load connected to **BUCK**.
 - Current of the load connected to **LVOUT**.
- The application circuit on **BUCK** does not pull current during the AEM0094x cold start.

To satisfy the last condition, the following circuit may be implemented:

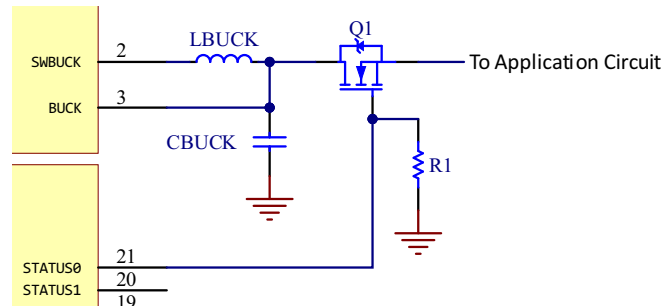


Figure 17: Schematic for supplying an application circuit with BUCK

Q1 is a N-MOSFET, whose gate is driven by **STATUS[0]** with R1 as a pull-down resistor. When the AEM0094x is in **DEEP SLEEP MODE** or in **WAKE-UP MODE**, **STATUS[0]** is LOW (see Section 5.2), ensuring that Q1 is non-conducting, and thus that the application circuit is not supplied.

When the AEM0094x switches from **WAKE-UP MODE** to **NORMAL MODE**, **STATUS[0]** is HIGH, making Q1 conducting. The application circuit is then supplied by **BUCK**, and remains so when the AEM0094x is in **NORMAL MODE**, **OVERCHARGE MODE**, **PRIMARY BATTERY MODE** and **SHUTDOWN MODE**.

Q1 must be chosen as follows:

- Low Gate-Source Leakage I_{GSS} .
- Low Zero Gate Voltage Drain Current I_{DSS} .
- Drain-Source On-State Resistance $R_{DS(on)}$ low enough to supply application circuit with an acceptable voltage drop.
- V_{GS} maximum voltage must be above V_{OVCH} (**STATUS[0]** HIGH voltage is V_{BOOST}).
- Maximum gate-source threshold voltage $V_{GS(th),MAX}$ matches the following, with $V_{BUCK,MAX}$ being V_{BUCK} maximum value stated in Table 6:

$$V_{GS(th),MAX} < V_{OVDIS} - V_{BUCK,MAX}$$

7. Typical Application Circuits

7.1. Example Circuit 1

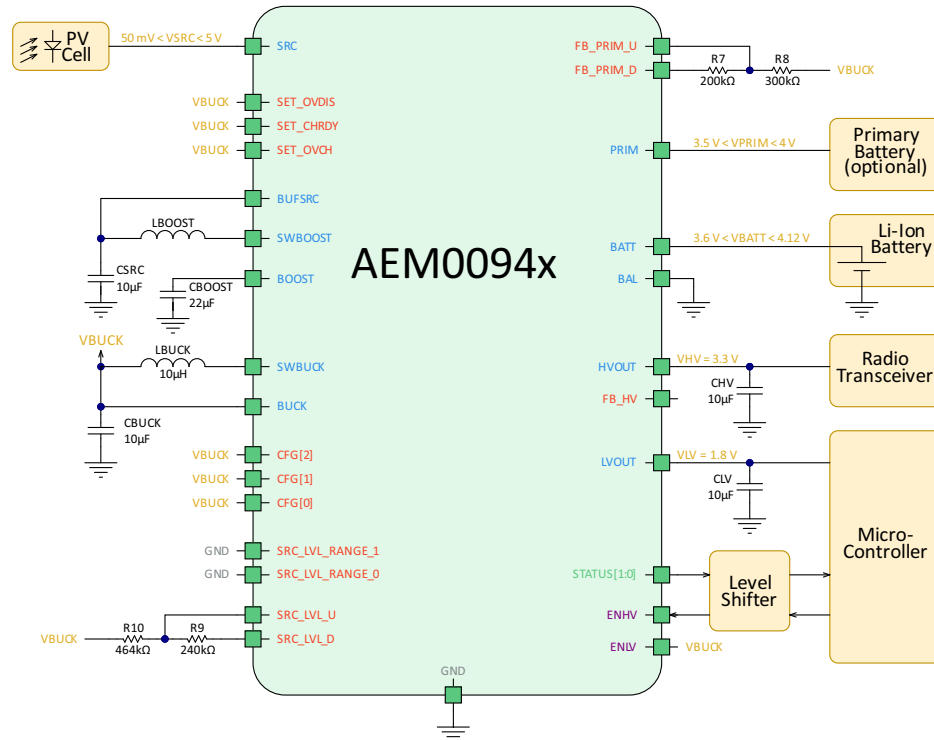


Figure 18: Typical application circuit 1

The energy source is a photovoltaic cell and the storage element is a standard Li-ion battery cell. The radio communication is supplied by **HVOUT** set at 3.3 V. The micro-controller that controls the application is supplied by **LVOUT** set at 1.8 V.

This circuit uses a pre-defined AEM configuration, typical of systems that use standard components for radio and energy storage.

The operating mode pins are set as follows:

- **CFG[2:0]** = HHH (all to **V_{BUCK}**)

Referring to Table 9, in this mode, the threshold voltages are:

- **V_{OVCH}** = 4.12 V
- **V_{CHRDY}** = 3.67 V
- **V_{OVDIS}** = 3.60 V

Moreover, the LDOs output voltages are:

- **V_{HV}** = 3.3 V
- **V_{LV}** = 1.8 V

A primary battery is also connected as a back-up solution. The minimal level allowed on this battery is set at 3.5 V. Following equations from Section 6.3:

- **R_p** = 0.5MΩ
- $R7 = \frac{3.5V}{4} \cdot 0.5M\Omega \cdot \frac{1}{2.2V} = 200k\Omega$
- **R8** = 0.5MΩ – 200kΩ = 300kΩ



The PV cell used needs a constant source voltage regulation at 0.75 V to extract the maximum power regardless of the brightness. **SRC_LVL_RANGE[1:0]** is set to LL according to Table 10.

- $R_S = 704\text{k}\Omega$
- $R_9 = \frac{0.75}{1} \cdot 704\text{k}\Omega \cdot \frac{1}{2.2\text{V}} = 240\text{k}\Omega$
- $R_{10} = 704\text{k}\Omega - 240\text{k}\Omega = 464\text{k}\Omega$

The **LVOUT** LDO output is enabled by tying **ENLV** to **V_{BUCK}**. The micro-controller is supplied by **LVOUT**, that is enabled when **V_{BATT}** and **V_{BOOST}** voltage rise above **V_{CHRDY}**.

The application software can enable or disable the radio transceiver supply with a GPIO connected to **ENHV**.

7.2. Example Circuit 2

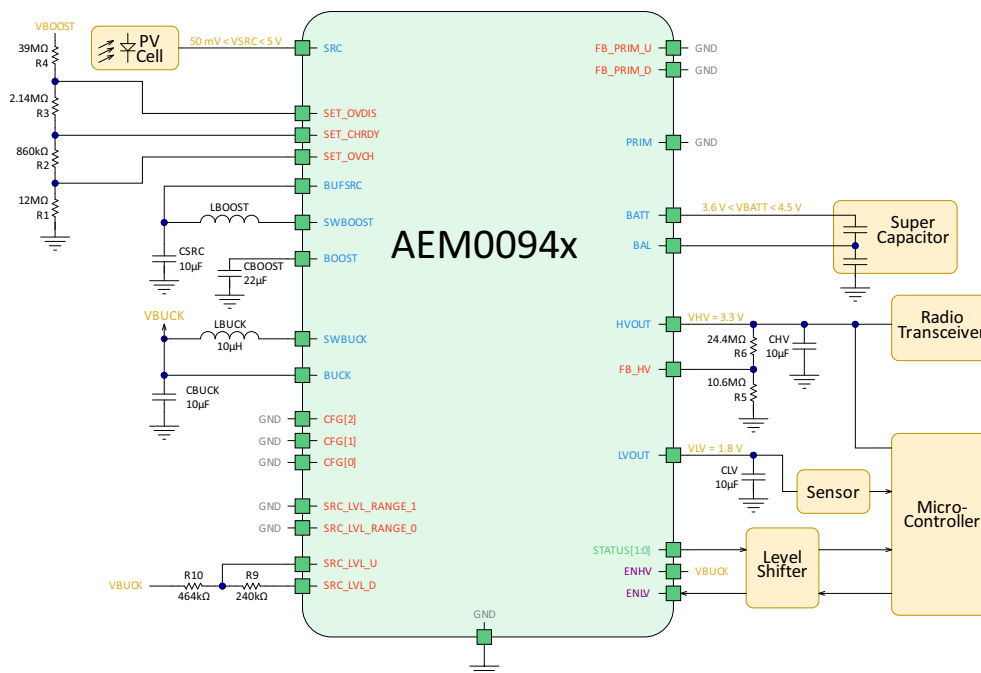


Figure 19: Typical application circuit 2

The energy source is a photovoltaic cell and the storage element is a dual-cell supercapacitor. Please note that the supercapacitor might be completely depleted during the cold start.

Moreover, **BAL** is connected to the dual-cell supercapacitor to compensate for any mismatch between the two cells and, in that way, protect the supercapacitor.

A micro-controller acts as the application master. The operating mode pins are set as follows:

- CFG[2:0] = LLL (all to GND)

The storage element voltages are set as follows with a custom configuration:

- $V_{OVCH} = 4.50 \text{ V}$
- $V_{CHRDY} = 4.20 \text{ V}$
- $V_{OVDIS} = 3.60 \text{ V}$

R_T is set to 54 M Ω . R_1 , R_2 , R_3 and R_4 values are computed from the equations in Section 6.1.1:

- $R1 = 54\text{M}\Omega \cdot \frac{1\text{V}}{4.5\text{V}} = 12\text{M}\Omega$
- $R2 = 54\text{M}\Omega \cdot \left(\frac{1\text{V}}{4.2\text{V}} - \frac{1\text{V}}{4.5\text{V}} \right) = 860\text{k}\Omega$
- $R3 = 54\text{M}\Omega \cdot \left(\frac{1\text{V}}{3.6\text{V}} - \frac{1\text{V}}{4.2\text{V}} \right) = 2.14\text{M}\Omega$
- $R4 = 54\text{M}\Omega \cdot \left(1 - \frac{1\text{V}}{3.6\text{V}} \right) = 39\text{M}\Omega$

The LDO voltages are set as follows:

- $V_{HV} = 3.3 \text{ V}$
- $V_{LV} = 1.8 \text{ V}$

LVOUT output enabling is controlled by the application circuit with a micro-controller GPIO connected to ENLV.



ENHV is tied to V_{BUCK} so that HVOUT is always on.

R_V is set to 35 M Ω . R5 and R6 are determined by applying the equations found in Section 6.1:

- $R5 = 35\text{M}\Omega \cdot \frac{1\text{V}}{3.3\text{V}} = 10.6\text{M}\Omega$
- $R6 = 35\text{M}\Omega \cdot \left(1 - \frac{1\text{V}}{3.3\text{V}}\right) = 24.4\text{M}\Omega$

The micro-controller is supplied by HVOUT, which is enabled when V_{BATT} and V_{BOOST} voltages rise above V_{CHRDY} .

The photovoltaic cell used needs a constant source voltage regulation at 0.75 V to provide the maximum power regardless of the ambient luminosity. SRC_LVL_RANGE[1:0] is set to LL according to the Table 10.

- $R_S = 704\text{k}\Omega$
- $R9 = \frac{0.75}{1} \cdot 704\text{k}\Omega \cdot \frac{1}{2.2\text{V}} = 240\text{k}\Omega$
- $R10 = 704\text{k}\Omega - 240\text{k}\Omega = 464\text{k}\Omega$

No primary battery is connected: PRIM, FB_PRIM_U and FB_PRIM_D pins are tied to GND.

8. Circuit Behavior

8.1. Cold-start Behavior with a (Super)Capacitor as Storage Element

8.1.1. Configuration

- **CFG[2:0]** = LHL
 - V_{OVCH} = 4.50 V
 - V_{CHRDY} = 3.92 V
 - V_{OVDIS} = 3.60 V
 - V_{HV} = 3.3 V
 - V_{LV} = 1.8 V
- C_{BATT} = 4.85 mF
- $V_{SRC,REG}$ = 2.1 V
 - **SRC_LVL_RANGE[1:0]** = LH (gain = 2)
 - R9 and R10 calculated so $V_{SRC_LVL_U}$ = 1.05 V
- **SRC**: 1 mA current source with 3 V voltage compliance
- **ENHV** = **ENLV** = H
- 22 k Ω resistive load on **LVOUT**
- 22 k Ω resistive load on **HVOUT**

8.1.2. Observations

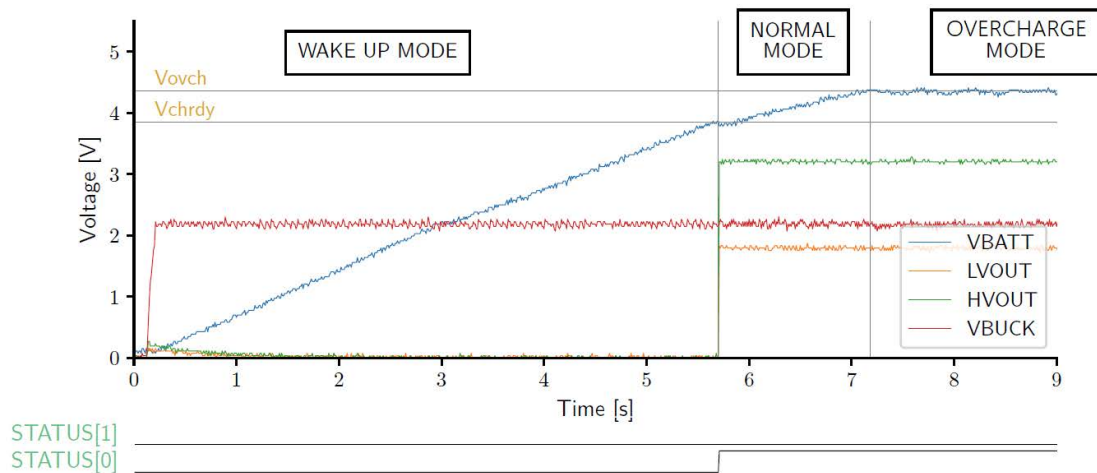


Figure 20: Cold start with a capacitor connected to BATT

- The AEM0094x is initially in **DEEP SLEEP MODE** with all nodes fully discharged. As soon as the current source is connected on **SRC**, the AEM0094x switches to **WAKE-UP MODE**, V_{BUCK} rises to its regulation voltage (around 2.2 V), and the storage element connected on **BATT** starts to be charged.
- Once V_{BATT} reaches V_{CHRDY} , the AEM0094x switches to **NORMAL MODE**, **STATUS[0]** is asserted and the LDOs are enabled. V_{HV} and V_{LV} are respectively regulated to 3.3 V and 1.8 V.
- Once V_{BATT} reaches V_{OVCH} , the AEM0094x switches to **OVERCHARGE MODE**. The voltage on **BATT** is maintained around V_{OVCH} . V_{BUCK} , V_{HV} and V_{LV} remain regulated.

8.2. Cold-start Behavior with a Battery as Storage Element

8.2.1. Configuration

- **CFG[2:0]** = HHH
 - **V_{OVCH}** = 4.12 V
 - **V_{CHRDY}** = 3.67 V
 - **V_{OVDIS}** = 3.60 V
 - **V_{HV}** = 3.3 V
 - **V_{LV}** = 1.8 V
- **C_{BATT}** = 4.85 mF (pre-charged capacitor acting as a battery)
- **V_{SRC,REG}** = 2.1 V
 - **SRC_LVL_RANGE[1:0]** = LH (gain = 2)
 - R9 and R10 calculated so **V_{SRC_LVL_U}** = 1.05 V
- **SRC**: 1 mA current source with 3 V voltage compliance
- **ENHV** = **ENLV** = H
- 22 kΩ resistive load on **LVOUT**
- 22 kΩ resistive load on **HVOUT**

8.2.2. Observations

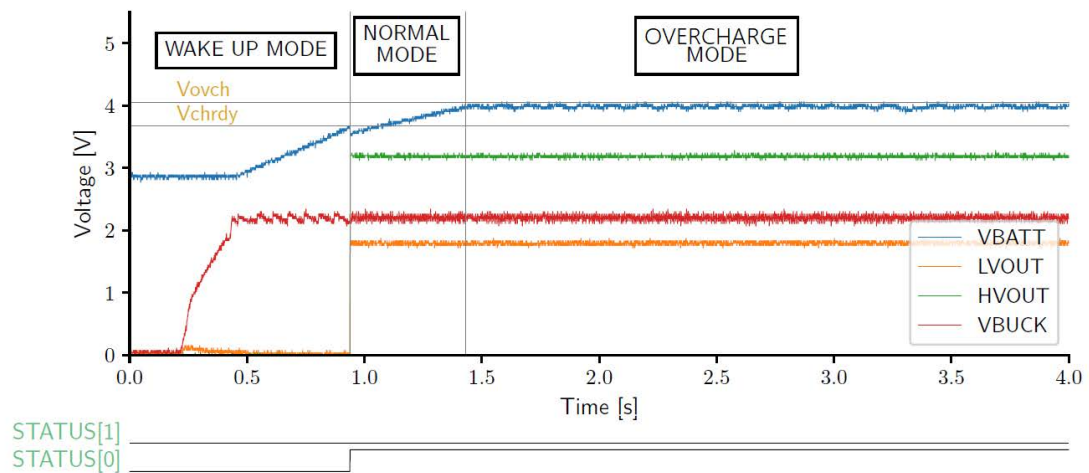


Figure 21: Cold start with a battery connected to BATT

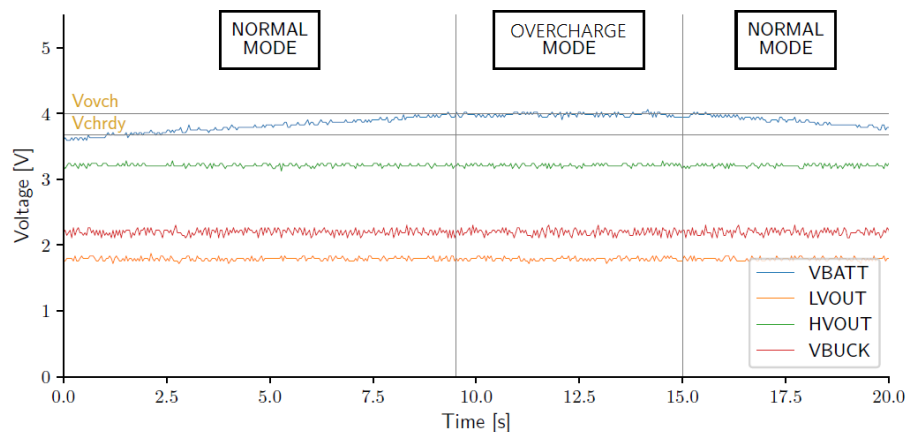
- The AEM0094x is initially in **DEEP SLEEP MODE** with all nodes fully discharged except **V_{BATT}** which is pre-charged. As soon as the current source is connected on **SRC**, the AEM0094x switches to **WAKE-UP MODE**, **V_{BUCK}** rises to its regulation voltage (around 2.2 V), and the storage element connected on **BATT** starts to be charged.
- Once **V_{BATT}** reaches **V_{CHRDY}**, the AEM0094x switches to **NORMAL MODE**, **STATUS[0]** is asserted and the LDOs are enabled. **V_{HV}** and **V_{LV}** are respectively regulated to 3.3 V and 1.8 V.
- Once **V_{BATT}** reaches **V_{OVCH}**, the AEM0094x switches to **OVERCHARGE MODE**. The voltage on **BATT** is maintained around **V_{OVCH}**. **V_{BUCK}**, **V_{HV}** and **V_{LV}** remain regulated.

8.3. Overcharge Mode Behavior

8.3.1. Configuration

- $CFG[2:0] = HHH$
 - $V_{OVCH} = 4.12\text{ V}$
 - $V_{CHRDY} = 3.67\text{ V}$
 - $V_{OVDIS} = 3.60\text{ V}$
 - $V_{HV} = 3.3\text{ V}$
 - $V_{LV} = 1.8\text{ V}$
- $C_{BATT} = 4.85\text{ mF}$
- $V_{SRC,REG} = 2.1\text{ V}$
 - $SRC_LVL_RANGE[1:0] = LH$ (gain = 2)
 - R9 and R10 calculated so $V_{SRC_LVL_U} = 1.05\text{ V}$
- SRC : 1 mA current source with 3 V voltage compliance
- $ENHV = ENLV = H$
- 22 k Ω resistive load on $LVOUT$
- 22 k Ω resistive load on $HVOUT$

8.3.2. Observations



STATUS[1]

STATUS[0]

Figure 22: Overcharge mode

- The AEM0094x is initially in **NORMAL MODE**, with $STATUS[0]$ asserted, and V_{BUCK} , V_{HV} and V_{LV} regulated.
- Thanks to the current source connected on SRC , V_{BATT} is being charged.
- Once V_{BATT} reaches V_{OVCH} , the AEM0094x enters **OVERCHARGE MODE** and maintain V_{BATT} around V_{OVCH} to avoid damaging the storage element.
- Near 15 s, the current source is disconnected from SRC . The AEM0094x maintain the two LDOs supplied, causing the storage element to discharge. The AEM0094x switches back to **NORMAL MODE**.



8.4. Shutdown Mode Behavior

8.4.1. Configuration

- $CFG[2:0] = LHL$
 - $V_{OVCH} = 4.50\text{ V}$
 - $V_{CHRDY} = 3.92\text{ V}$
 - $V_{OVDIS} = 3.60\text{ V}$
 - $V_{HV} = 3.3\text{ V}$
 - $V_{LV} = 1.8\text{ V}$
- $C_{BATT} = 4.85\text{ mF}$
- $V_{SRC,REG} = 2.1\text{ V}$
 - $SRC_LVL_RANGE[1:0] = LH$ (gain = 2)
 - R9 and R10 calculated so $V_{SRC_LVL_U} = 1.05\text{ V}$
- SRC : left floating to let the storage element on $BATT$ discharge
- $ENHV = ENLV = H$
- $22\text{ k}\Omega$ resistive load on $LVOUT$
- $22\text{ k}\Omega$ resistive load on $HVOUT$
- $PRIM$, FB_PRIM_U and FB_PRIM_D connected to GND (no primary battery)

8.4.2. Observations

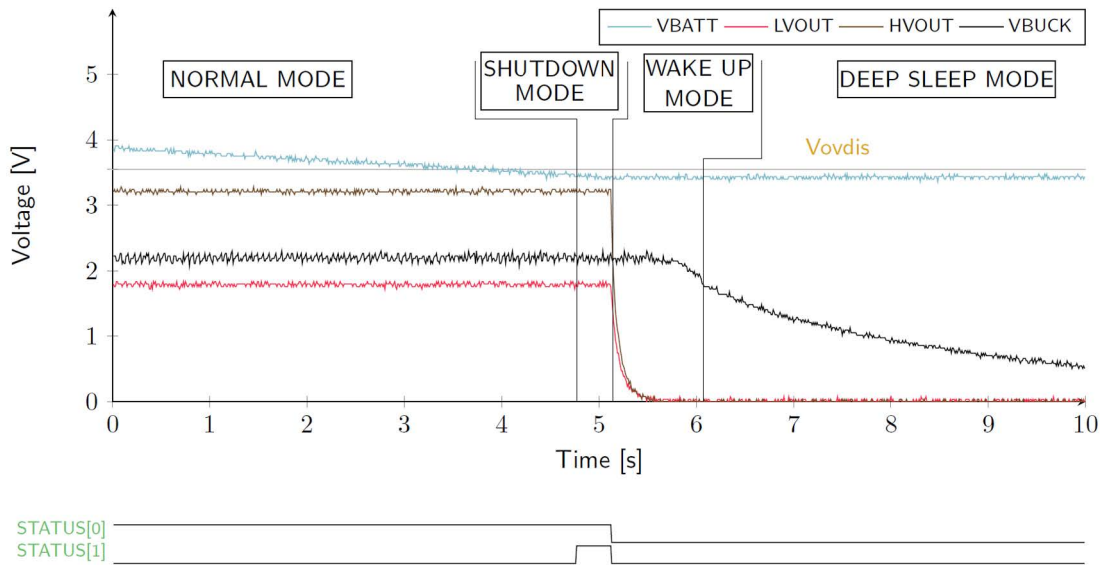


Figure 23: Shutdown mode (without primary battery)

- The AEM0094x is initially in **NORMAL MODE**, with **STATUS[0]** asserted, V_{BUCK} , V_{HV} and V_{LV} regulated. The **SRC** pin is left floating, which result in the storage element discharging through the loads.
- Once V_{BATT} reaches V_{OVDIS} , the AEM0094x enters **SHUTDOWN MODE** and maintain the load outputs regulated. **STATUS[1]** is asserted to warn the user that a shutdown may occur.
- After T_{CRIT} (approximately 600 ms), V_{BATT} has not recovered above V_{OVDIS} . The AEM0094x disables the LDOs and enters **WAKE-UP MODE**. To avoid further discharging the storage element, **BATT** is disconnected from **BOOST**. The pins **STATUS[1]** and **STATUS[0]** are set LOW.
- Near 6.1 s, V_{BUCK} falls below $V_{BUCK,RESET}$. The AEM0094x enters **DEEP SLEEP MODE** and must go through the cold-start procedure again.



8.5. Primary Battery Mode Behavior

8.5.1. Configuration

- $CFG[2:0] = HHH$
 - $V_{OVCH} = 4.12\text{ V}$
 - $V_{CHRDY} = 3.67\text{ V}$
 - $V_{OVDIS} = 3.60\text{ V}$
 - $V_{HV} = 3.3\text{ V}$
 - $V_{LV} = 1.8\text{ V}$
- $C_{BATT} = 4.85\text{ mF}$
- $V_{SRC,REG} = 2.1\text{ V}$
 - $SRC_LVL_RANGE[1:0] = LH$ (gain = 2)
 - R9 and R10 calculated so $V_{SRC_LVL_U} = 1.05\text{ V}$
- SRC : 1 mA current source with 3 V voltage compliance (initially disconnected)
- $ENHV = ENLV = H$
- 22 k Ω resistive load on $LVOUT$
- 22 k Ω resistive load on $HVOUT$
- $PRIM$: 3 V voltage source with 1 mA current compliance
 - $R7 = 68\text{ k}\Omega$
 - $R8 = 330\text{ k}\Omega$
 - $V_{PRIM,MIN} = 1.50\text{ V}$

8.5.2. Observations

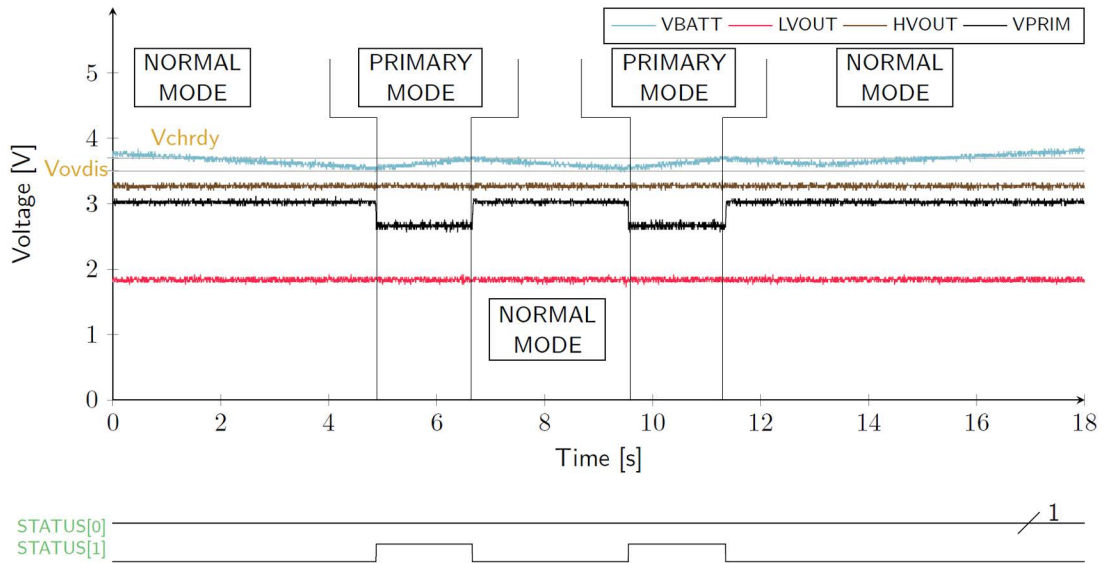


Figure 24: Switching to primary battery when battery is overdischarged

- The AEM0094x is initially in **NORMAL MODE**, with **STATUS[0]** asserted, V_{BUCK} , V_{HV} and V_{LV} regulated.
- The **SRC** is left floating, causing the storage element to discharge.
- Once V_{BATT} reaches V_{OVDIS} , the AEM0094x compares the voltage on **PRIM** to $V_{PRIM,MIN}$. Since V_{PRIM} is higher, the AEM enters **PRIMARY BATTERY MODE** and, the primary battery is used to recharge the storage element until V_{BATT} reaches V_{CHRDY} again.
- **STATUS[1]** is asserted as long as the AEM0094x is in **PRIMARY BATTERY MODE**.
- Once V_{BATT} reaches V_{CHRDY} , the AEM0094x goes back into **NORMAL MODE**. Since the **SRC** is still floating, the same behavior repeats.
- Near 13 s, while the storage element is discharging in **NORMAL MODE**, the current source is connected on **SRC**. This allows charging the storage element so the primary battery is not used anymore.



9. Schematic

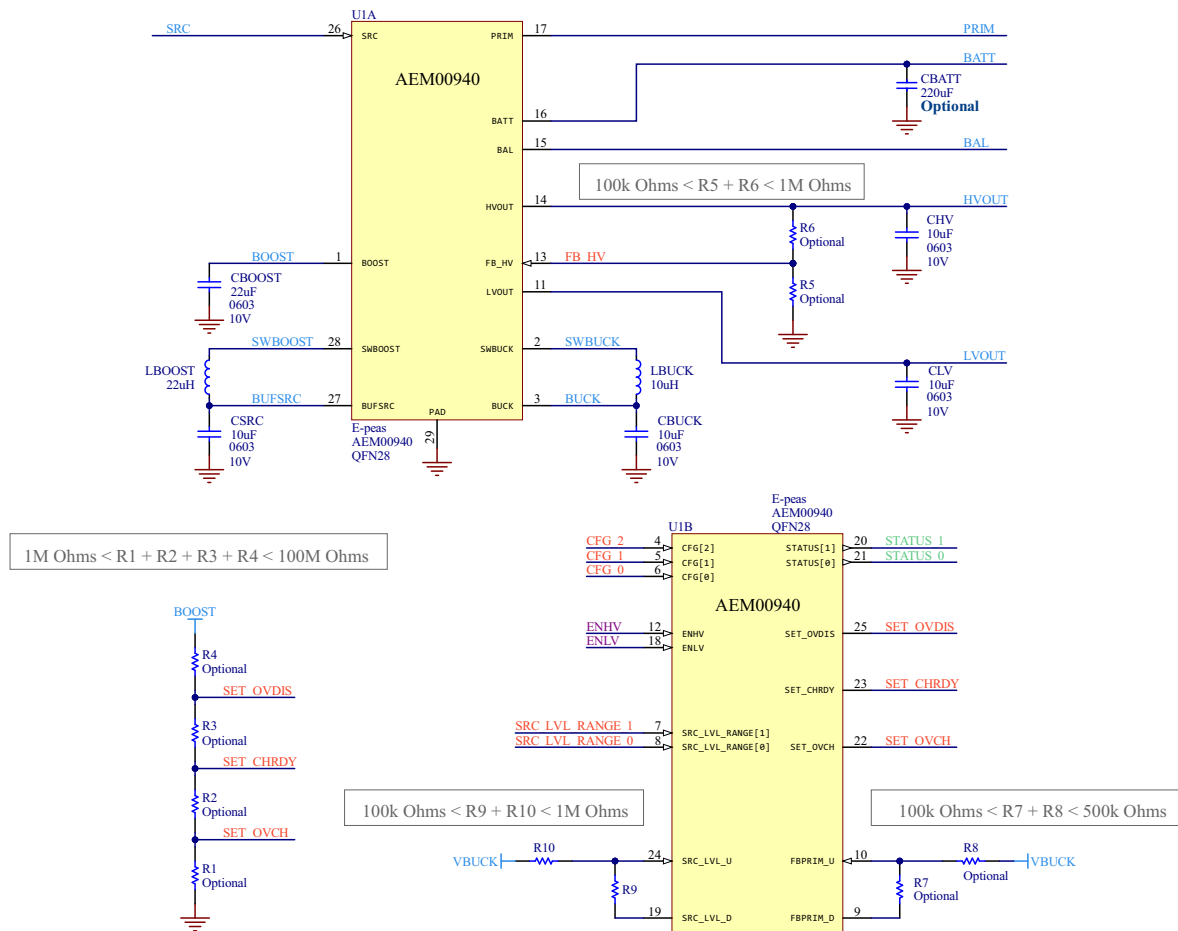


Figure 25: Schematic example

Designator	Description	Qty	Manufact.	Part Number
U1	AEM0094x, QFN28	1	e-peas	order at sales@e-peas.com
C _{SRC}	Ceramic Cap 10 µF, 16 V, 20 %, X5R, 0603	1	Murata	GRM188R61C106MA
C _{BOOST}	Ceramic Cap 22 µF, 10 V, 20 %, X5R, 0603	1	Murata	GRM188R61A226ME
For AEM00940: L _{BOOST} L _{BOOST} (alt.) L _{BOOST} (alt.)	Power Inductor 22 µH, 0.65 A, LPS4018 Power Inductor 10 µH, 0.90 A, LPS4018 Power Inductor 10 µH, 0.84 A, 3015	1 1 1	Coilcraft Coilcraft Würth	LPS4018-223MR LPS4018-103MR 744 040 321 00
For AEM00941: L _{BOOST}	Power Inductor 100 µH, 0.55 A, LPS5030	1	Coilcraft	LPS5030-104MR
C _{BUCK}	Ceramic Cap 10 µF, 16 V, 20 %, X5R, 0603	1	Murata	GRM188R61C106MA
L _{BUCK}	Power Inductor 10 µH, 0.25 A, 0603	1	TDK	MLZ1608N100LT000
C _{LV}	Ceramic Cap 10 µF, 16 V, 20 %, X5R, 0603	1	Murata	GRM188R61C106MA
C _{HV}	Ceramic Cap 10 µF, 16 V, 20 %, X5R, 0603	1	Murata	GRM188R61C106MA
C _{BATT}	Ceramic Cap 220 µF, 6.3 V, 20 %, X5R, 1206	1	Murata	GRM31CR60J227ME

Table 11: BOM example for AEM0094x and its required passive components



10. Layout

10.1. Guidelines

Good layout practices are mandatory in order to obtain good AEM0094x stability, best efficiency and avoid EMI problems.

The following list, while not exhaustive, shows the main attention points when routing a PCB with the AEM0094x:

- The switching nodes (**SWBOOST** and **SWBUCK**) must be kept as short as possible, with minimal track resistance and minimal track capacitance. Low resistance is obtained by keeping track length as short as possible and track width as large as possible between these switching nodes and the AEM0094x pins. Minimal capacitance is obtained by maintaining a large distance between the switching nodes and other signals. We recommend removing the ground plane, the power plane and the bottom layer ground pour under **L_{BOOST}** and **L_{BUCK}** footprints, as well as adding distance between **BUFSRC/SWBOOST** and the top ground pour, as shown in Figure 26.
- The decoupling capacitors (**C_{BOOST}** - **C_{BUCK}** - **C_{SRC}** - **C_{HV}** - **C_{LV}** - **C_{BATT}**) must be placed as close as possible to the AEM0094x, with direct connection and minimum track resistance for the corresponding power nodes (**BOOST**, **BUCK**, **BUFSRC**, **HVOUT**, **LVOUT** and **BATT**).
- The **GND** return path between the decoupling capacitors and the AEM0094x thermal pad, which is the AEM0094x main **GND** connection, must be as direct and short as possible. This is preferably done on the top layer when possible, otherwise by internal/bottom plane, using low resistance vias to decrease layer-to-layer connection resistance. In Figure 26, this **GND** return path is done on an internal plane.
- The external DC power connections (**SRC**, **HVOUT**, **LVOUT** and **BATT**) must be connected to the AEM0094x with low resistance tracks.
- The **BAL** pin connection track must be able to handle at least 40 mA.
- The custom mode setting pins **SET_OVDIS**, **SET_CHRDY** and **SET_OVCH** are high impedance analog inputs typically connected to a resistive divider with high resistor values, making those three nodes prone to pickup noise. Thus, it is recommended to keep those as short as possible and as far as possible to noise sources such as DCDC switching nodes.
- The configuration pins and the status pins have minimal layout restrictions.

10.2. Layout Example

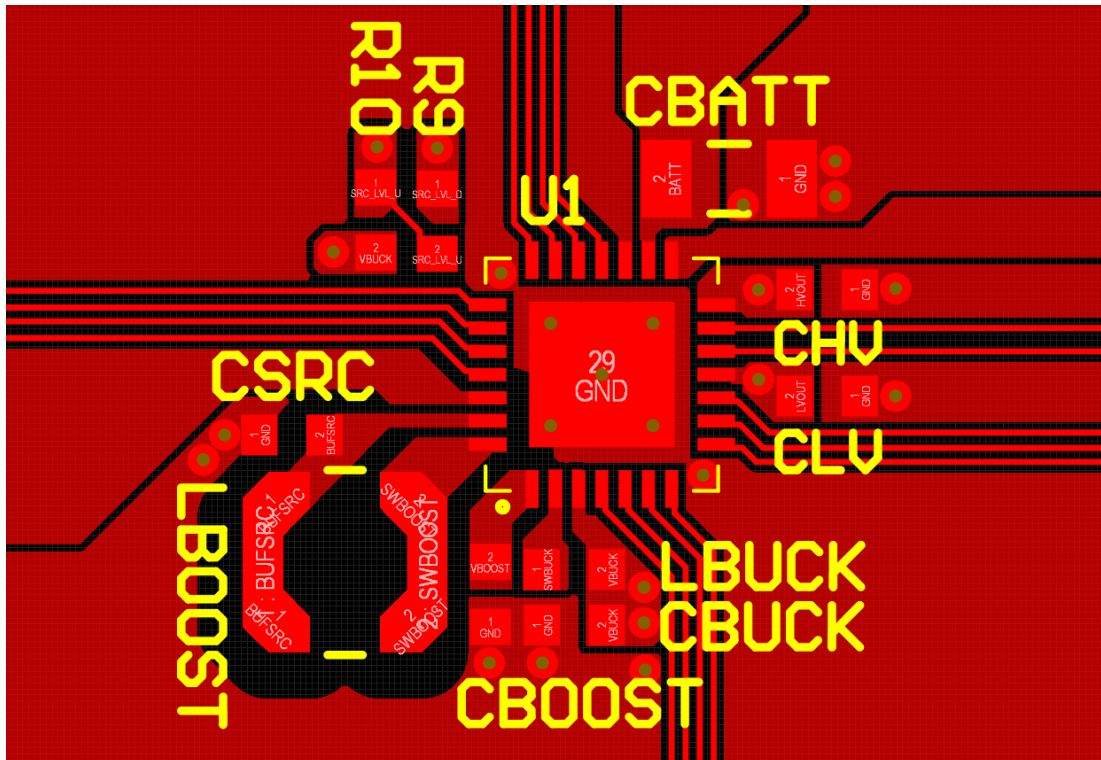


Figure 26: Layout example for the AEM00940 and its passive components

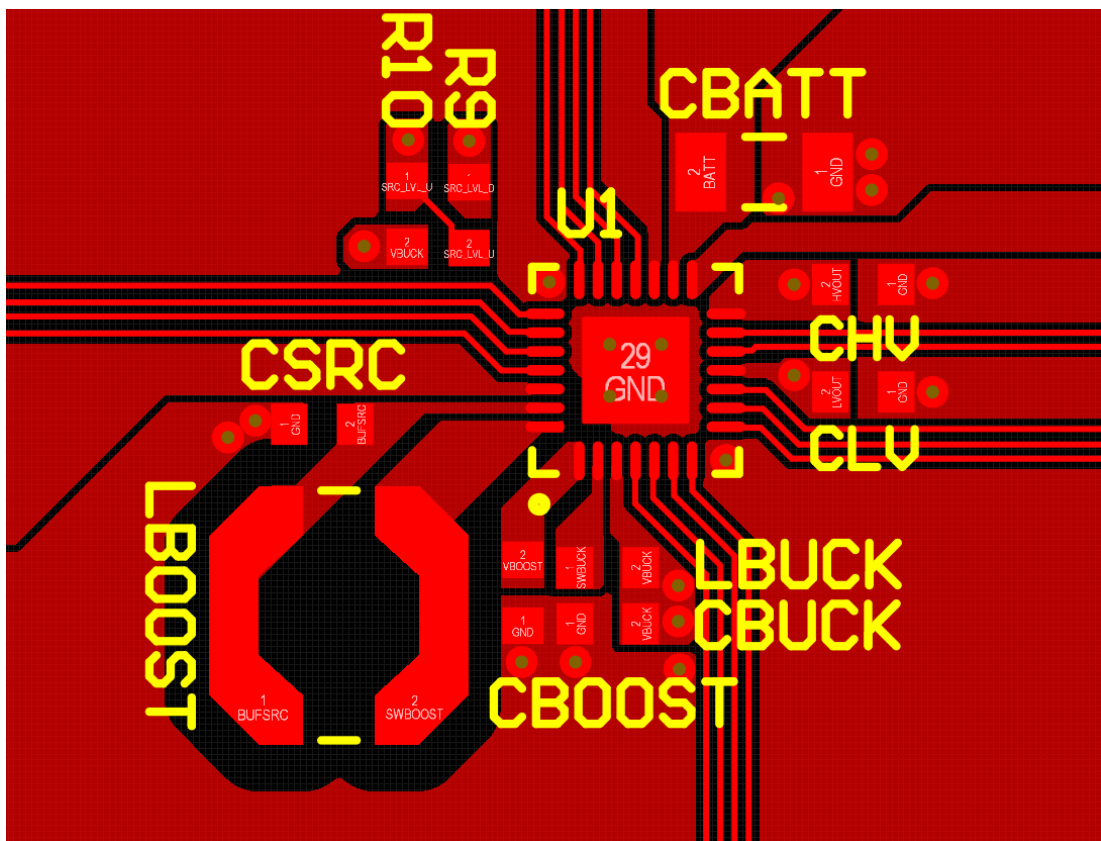


Figure 27: Layout example for the AEM00941 and its passive components

11. Package Information

11.1. Moisture Sensitivity Level

Package	Parameter	Value
QFN 28-pin 5x5mm	Moisture Sensitivity Level (MSL) ¹	Level 1
QFN 28-pin 4 x 4 mm	Moisture Sensitivity Level (MSL) ¹	Level 1

Table 12: Moisture sensitivity level

1. According to JEDEC 22-A113 standard.

11.2. RoHS Compliance

e-peas product complies with RoHS requirement.

e-peas defines “RoHS” to mean that semiconductor end-products are compliant with RoHS regulation for all 10 RoHS substances.

This applies to silicon, die attached adhesive, gold wire bonding, lead frames, mold compound, and lead finish (pure tin).

11.3. REACH Compliance

The component and elements used by e-peas subcontractors to manufacture e-peas PMICs and devices are REACH compliant. For more detailed information, please contact e-peas sales team.

11.4. Tape and Reel Dimensions

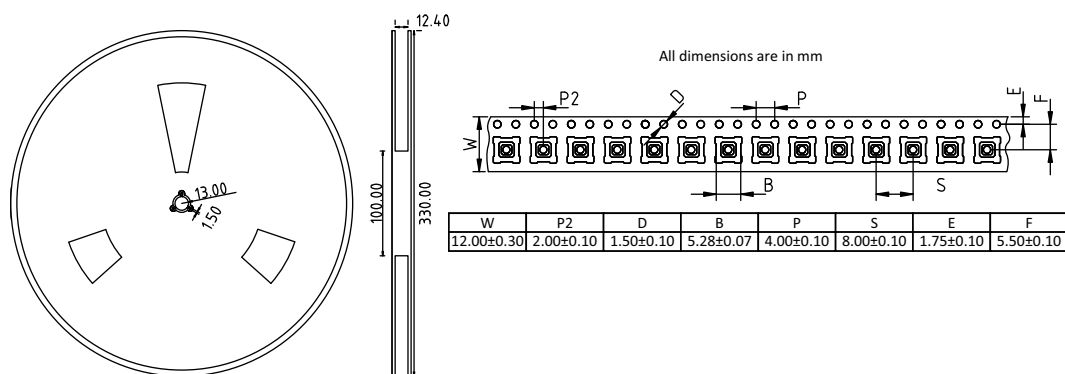


Figure 28: Tape and reel dimensions

11.5. Package Dimensions

11.5.1. AEM00940 QFN 28-pin 5x5mm

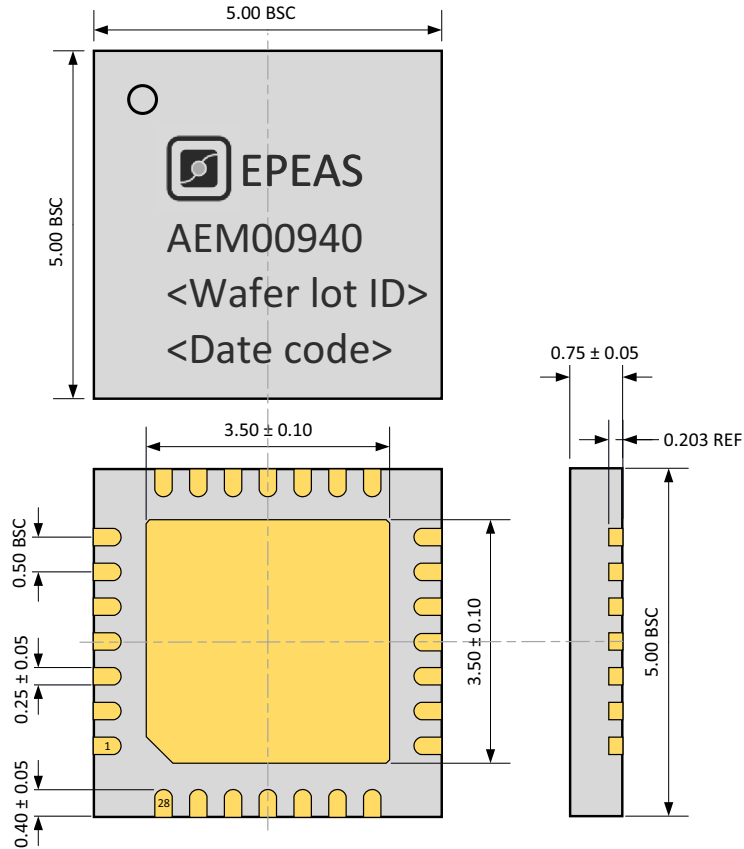


Figure 29: AEM00940 QFN 28-pin 5x5mm drawing (all dimension in mm)



11.5.2. AEM00941 QFN 28-pin 4x4mm

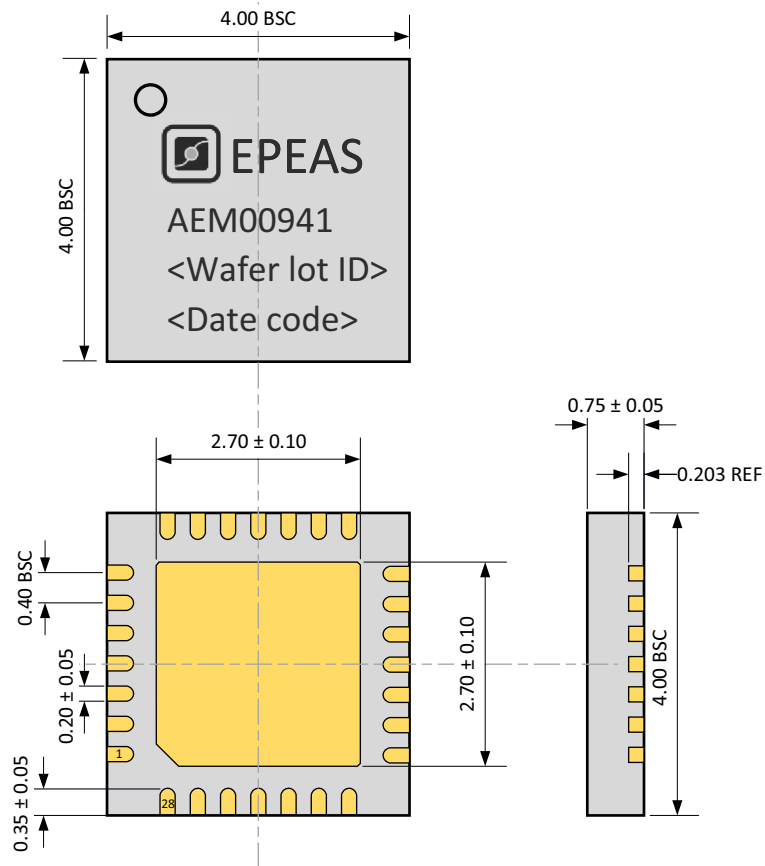


Figure 30: AEM00941 QFN 28-pin 4x4mm drawing (all dimension in mm)

11.6. Board Layout

11.6.1. AEM00940 QFN 28-pin 5x5mm

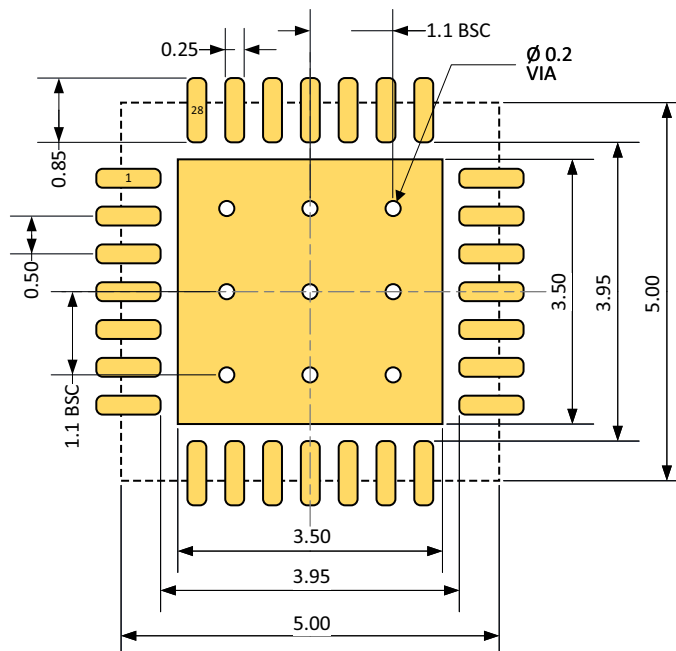


Figure 31: Recommended board layout for QFN 28-pin 5x5mm (all dimension in mm)

11.6.2. AEM0091 QFN 28-pin 4x4mm

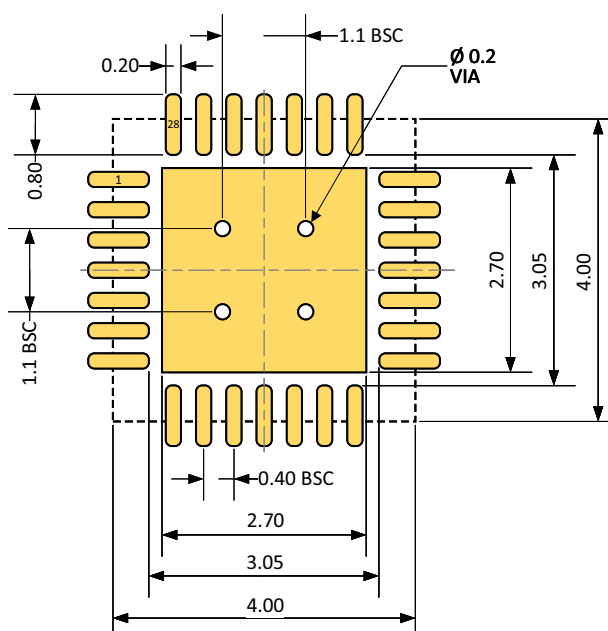


Figure 32: Recommended board layout for QFN 28-pin 4x4mm (all dimension in mm)



12. Glossary

Symbol	Description
AEM	Ambient Energy Manager.
BOM	Bill Of Materials.
C_{BATT}	Capacitor connected on the BATT pin.
C_{BOOST}	Output capacitor of the BOOST converter.
C_{BUCK}	Output capacitor of the BUCK converter.
C_{HV}	High-voltage LDO regulator decoupling capacitor.
C_{LV}	Low-voltage LDO regulator decoupling capacitor.
C_{SRC}	BUFSRC pin decoupling capacitor.
GPIO	General Purpose Input / Output.
I_{BUCK}	Total load current supplied by the BUCK converter (including the LVOUT current I_{LV}).
I_{HV}	Load current supplied by the high-voltage LDO regulator.
I_{LV}	Load current supplied by the low-voltage LDO regulator.
I_{PRIM}	Current from the primary battery.
I_Q	Quiescent current on BATT when no energy is available on SRC .
I_{SRC}	Harvested current from the energy source.
L_{BOOST}	BOOST converter inductor.
L_{BUCK}	BUCK converter inductor.
LDO	Low Drop-Out linear regulator.
PCB	Printed Circuit Board.
$P_{SRC,CS}$	Minimum power on SRC for the AEM0094x to coldstart.
R_P	Sum of resistors for setting the primary battery minimum voltage. $R_P = R7 + R8$.
R_S	Sum of resistors for setting the source voltage regulation. $R_S = R9 + R10$.
R_T	Sum of resistors for setting the battery protection threshold voltages in custom mode. $R_T = R1 + R2 + R3 + R4$.
R_V	Sum of resistors for setting the output voltage of the high-voltage LDO in custom mode. $R_V = R5 + R6$.
T_{CRIT}	Time during which the AEM stays in SHUTDOWN MODE before returning to NORMAL MODE or going to DEEP SLEEP MODE .
$T_{VSRC,REG,UPDATE}$	Time between two updates of the source voltage regulation.
V_{BAL}	Voltage on the BAL pin.
$V_{BAL,MIN}$	Minimum voltage on the BAL pin for the balancing circuit to operate.
V_{BATT}	Voltage on the BATT pin.
V_{BOOST}	Output voltage of the BOOST converter.
V_{BUCK}	Output voltage of the BUCK converter.
$V_{BUCK,RESET}$	Minimum voltage on BUCK before switching to DEEP SLEEP MODE (from any other mode).
V_{CHRDY}	Charge ready voltage on the BATT pin.
$V_{FB_PRIM_U}$	Feedback for the minimal voltage level on the primary battery.
V_{HV}	Output voltage of the high-voltage LDO regulator.
V_{LV}	Output voltage of the low-voltage LDO regulator.

Table 13: Glossary (Part 1)



Symbol	Description
V_{OC}	Open-circuit voltage of the harvester connected to the SRC pin.
V_{OVCH}	Over-charge voltage on the BATT pin.
V_{OVDIS}	Over-discharge voltage on the BATT pin.
V_{PRIM}	Voltage on the primary battery.
$V_{PRIM,MIN}$	Voltage at which the primary battery is considered fully depleted.
V_{SRC}	Voltage on the SRC pin.
$V_{SRC,CS}$	Minimum voltage on SRC for the AEM0094x to coldstart.
$V_{SRC_LVL_U}$	Voltage on the SRC_LVL_U pin.
$V_{SRC,REG}$	Target regulation voltage on the SRC pin.

Table 13: Glossary (Part 2)



13. Revision History

Revision	Date	Description
1.2	April, 2026	- Document structure updated (section order and titles revised). - STATUS[1] description modified in pin description table. - Updated the maximum operating junction temperature to 85°C. - Updated HBM ESD value. - Updated the thermal resistance for QFN 28-pin 4 x 4mm. - Addition of REACH and RoHS compliance statements. - VPRIM maximum value modified for VOVDIS + 0.4 V. Corresponding explanations added in the Boost Converter section. - VBUCK,RESET and VBAL,MIN values added in Electrical Characteristics table. - Diagram of the AEM0094x modes modified with new links between modes. - Primary Battery Mode section, Shutdown Mode section, and Storage Element Balancing Circuit section reworked. - Addition of a disclaimer about the battery threshold levels. - Reworked and moved the external inductors, external capacitors, and storage element information sections. - Circuit Behavior sections rework. - Change LBOOST recommenced value to 22 µH for best efficiency throughout the datasheet. - Change CBATT recommended value to 220 µF in “Schematic” section. - Updated LBUCK part number in the BOM example table. - Updated VPRIM maximum value in the Simplified schematic view and Typical application circuit 1 figures. - Added moisture sensitivity level and tape and reel dimensions in the package information section. - Updated the package dimensions and the recommended board layouts.

Table 14: Revision history (Part 1)



Revision	Date	Description
1.1	January, 2024	- AEM00941 added. AEM0094x used when both AEM00940 and AEM00941 concerned. - R9 and R10 swapped in the whole document. - SET_OVDIS, SET_CHRDY and SET_OVCH pins changed from “left floating” to “connected to VBUCK” when not used (Table 1 and Section 6.1). - Added “Pin Configuration and Functions” section containing the pinout diagram and description. - Electrical characteristics table: - VSRC maximum value during and after cold start modified to 5 V. - ISRC value added for LBOOST values of 22 μH and 100 μH. - Different min values for VBATT depending on used storage element removed. - VPRIM maximum value changed from 4.5 V to 5 V. - VHV maximum value changed from “VBATT - 0.3 V” to “VOVDIS - 0.3 V”. - Added IBUCK, TVSRC_REG.UPDATE and IQ values. - Recommended operating conditions table: - CSRC maximum value of 22 μF added. - LBOOST value of 100 μH added for the AEM00941. - Inductors unit error corrected to μH. - CBOOST, CBUCK and LBOOST maximum values removed. - ENHV minimum and maximum values removed. - ENLV minimum value replaced by VBUCK. - Condition to go from SHUTDOWN mode to NORMAL mode corrected to “If VBATT > VOVDIS” in Figure 14 and Section 5.2.5. - Added the Source Voltage Regulation section. - Added how to use BUCK to supply an application, and BUCK converter performances. - Second CBATT value added of 22 μF when storage element not connected and LDOs not used (Section 6.6). - Modified VOVDIS, R2, R3 and R4 values in Example Circuit 2. - New “Circuit Behavior” section created to place the circuit behavior figures. - SHUTDOWN mode behavior figures (with and without primary battery) added. - Added AEM00941 efficiency graphs with LBOOST = 100 μH. - BOM and required passive components example added. - Added layout guidelines section and glossary section. - QFN-28pin 4x4mm package information added for AEM00941. - Replace 0 by L for logic low and 1 by H for logic high in tables and texts.
1.0	February, 2022	Creation of the document.

Table 14: Revision history (Part 2)